

GREAT VESSELS OF THE NECK

Common Carotid Artery (CCA)

Origin:

- **Right artery:** Arises in the neck from the brachiocephalic artery behind the right sternoclavicular joint.
- **Left artery:** Arises in the thorax from the arch of the aorta, behind manubrium sterni.

Course and termination:

It ascends inside the carotid sheath medial to vagus nerve and internal jugular vein

ends at the upper border of thyroid cartilage (at disc between C3 and C4 vertebrae) by dividing into external carotid artery (E.C.A.) and internal carotid artery (I.C.A.).

Relations:

Anteriorly:

- **Muscles:** Infrahyoid and sternomastoid muscles.
- **Gland:** Thyroid lobe.

Posteriorly:

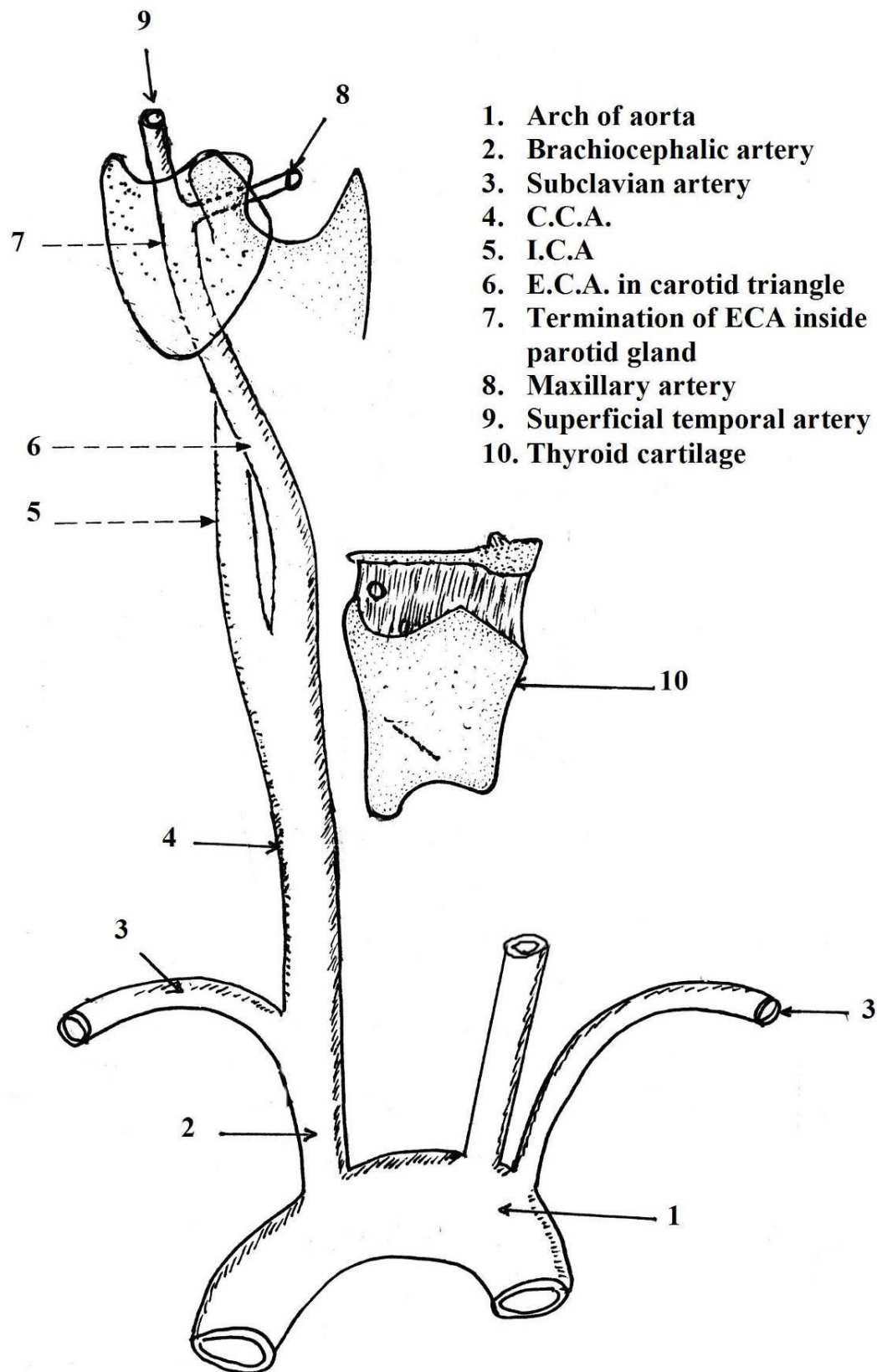
- **Cervical vertebrae** and prevertebral muscles.
- **Arteries:** Inferior thyroid artery, vertebral artery.
- **Nerves:** Sympathetic trunk and right recurrent laryngeal nerve.

Medially:

- **Larynx and pharynx** (above).
- **Trachea and oesophagus** with recurrent laryngeal nerve in between (below).
- **Laterally:** internal jugular vein.

Surface anatomy of the carotid arteries:

A line drawn from the sternoclavicular joint to a point midway between the angle of mandible and the mastoid process. This line represents CCA and ICA and a point in this line corresponding to the upper border of thyroid cartilage is the junction between them.



Great Vessels in the Head & Neck

External Carotid Artery (E.C.A.)

Origin:

It arises from CCA at the upper border of thyroid cartilage (at disc between C3 and C4 vertebrae).

Course and termination:

It ascends anteromedial to internal carotid artery outside carotid sheath and ends behind the neck of mandible in the substance of parotid gland by dividing into:

superficial temporal and maxillary arteries.

Branches of external carotid artery:

1- Three from the anterior aspect: Superior thyroid, lingual and facial arteries.

2- Two from the posterior aspect: Occipital and posterior auricular arteries.

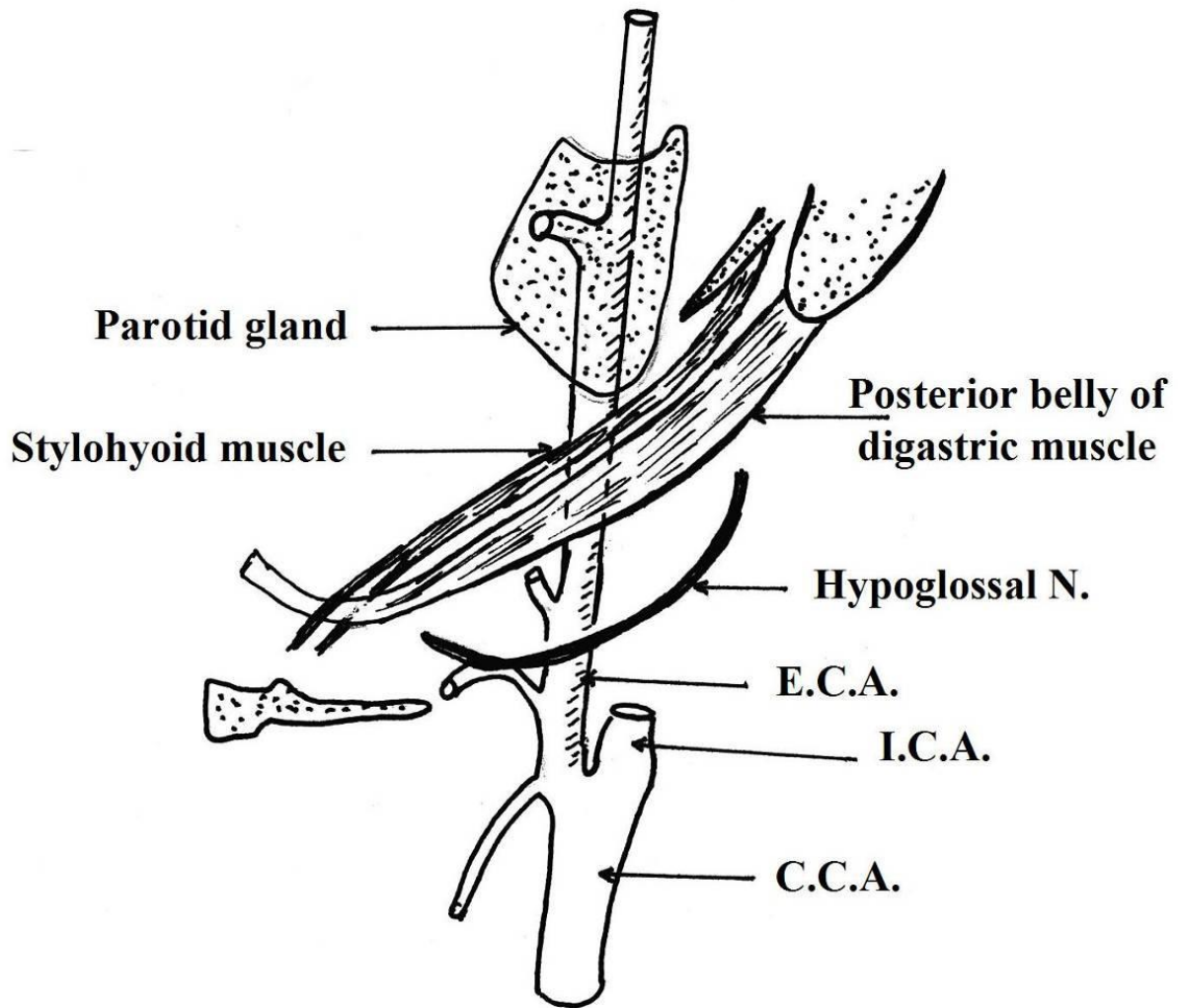
3- One from the medial (deep) aspect: Ascending pharyngeal artery.

4- Two terminal branches: Within the substance of parotid gland: maxillary and superficial temporal arteries.

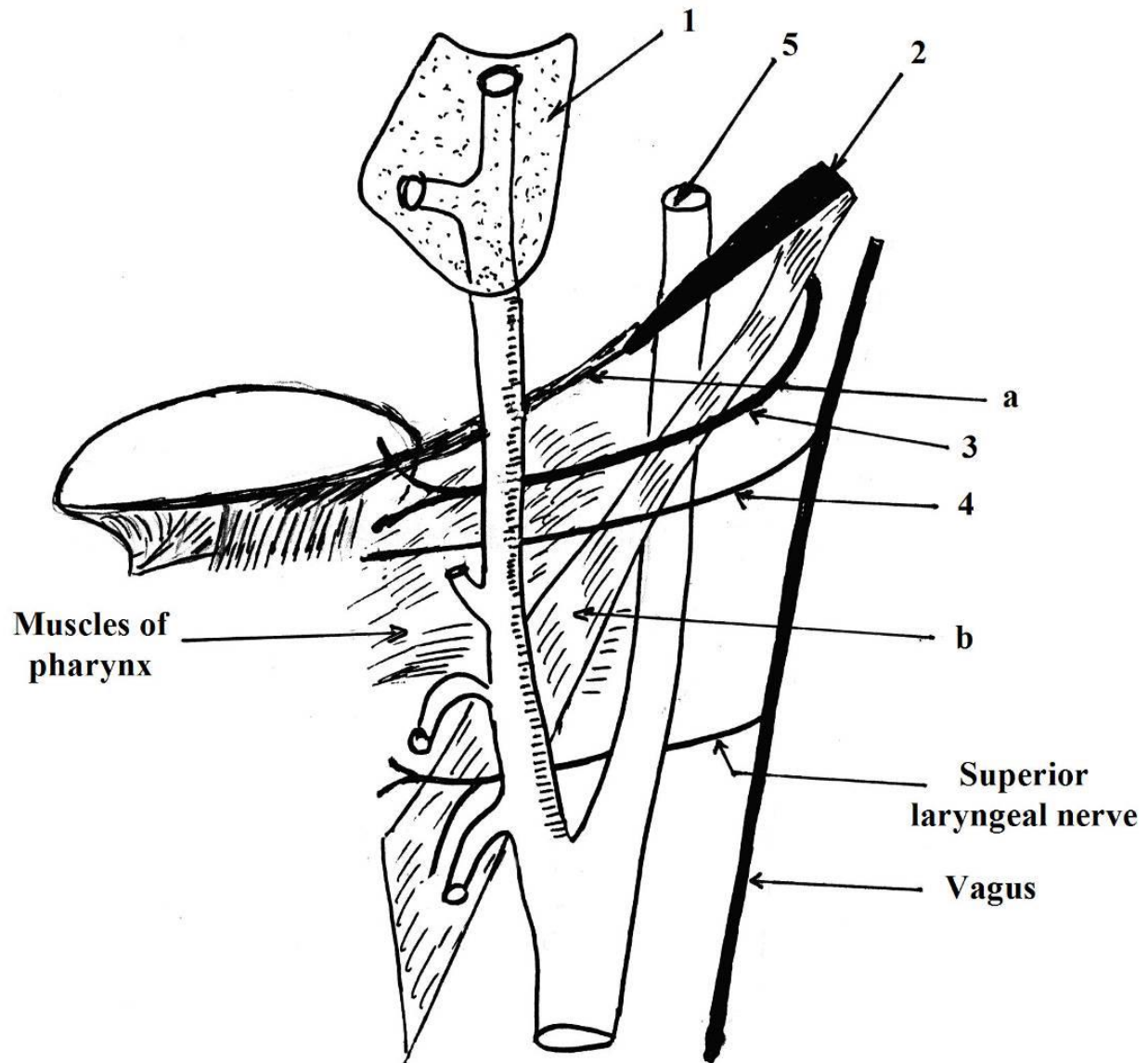
Relations:

A- Superficial relations (3 muscles, 2 veins, one nerve, one gland):

- **Three muscles:** Sternomastoid, posterior belly of digastric and stylohyoid muscles.
- **Two veins:** Lingual and common facial veins.
- **One nerve:** Hypoglossal nerve.
- **Parotid gland:** With the retromandibular vein inside the gland.



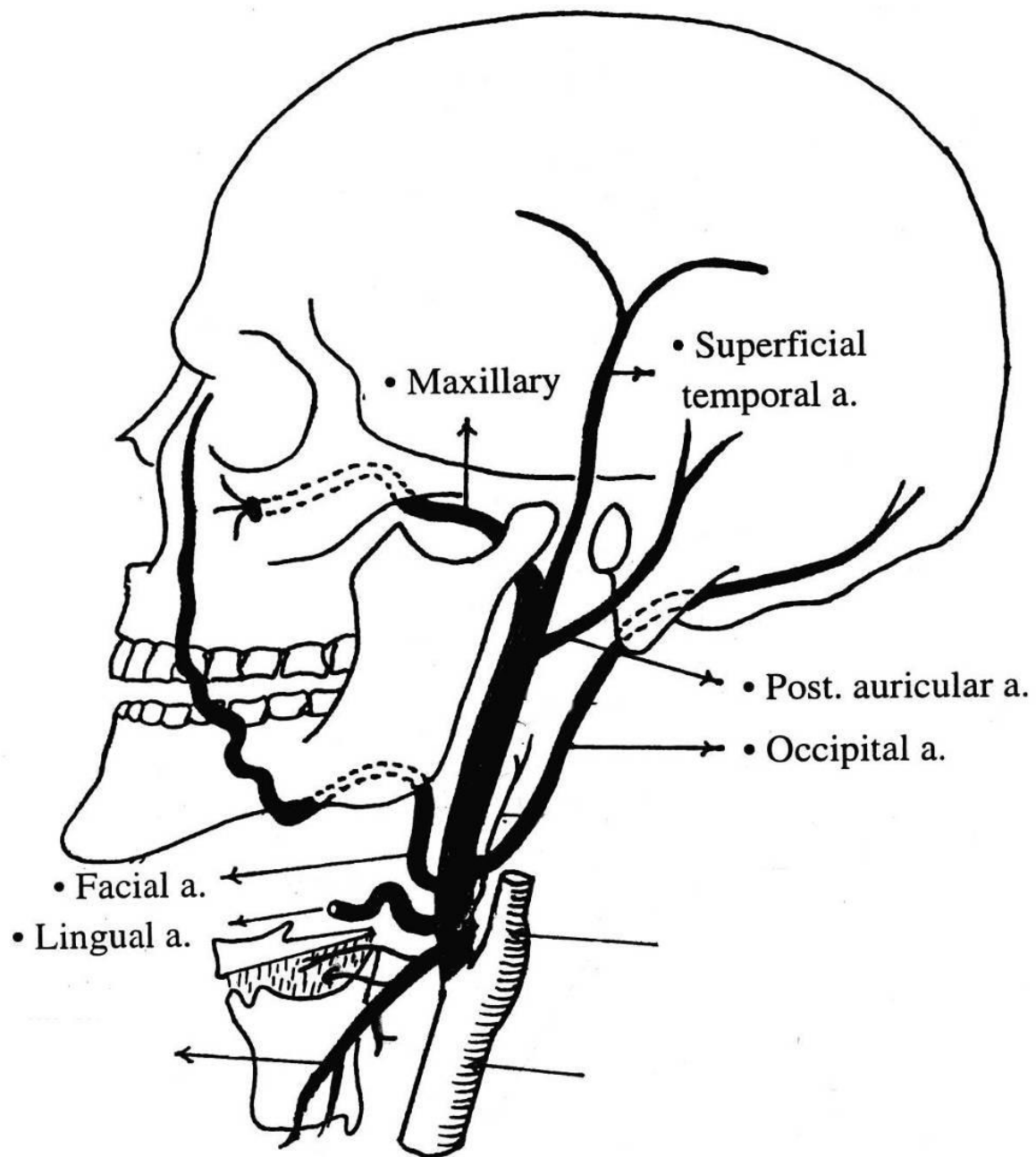
Superficial Relations of E.C.A.



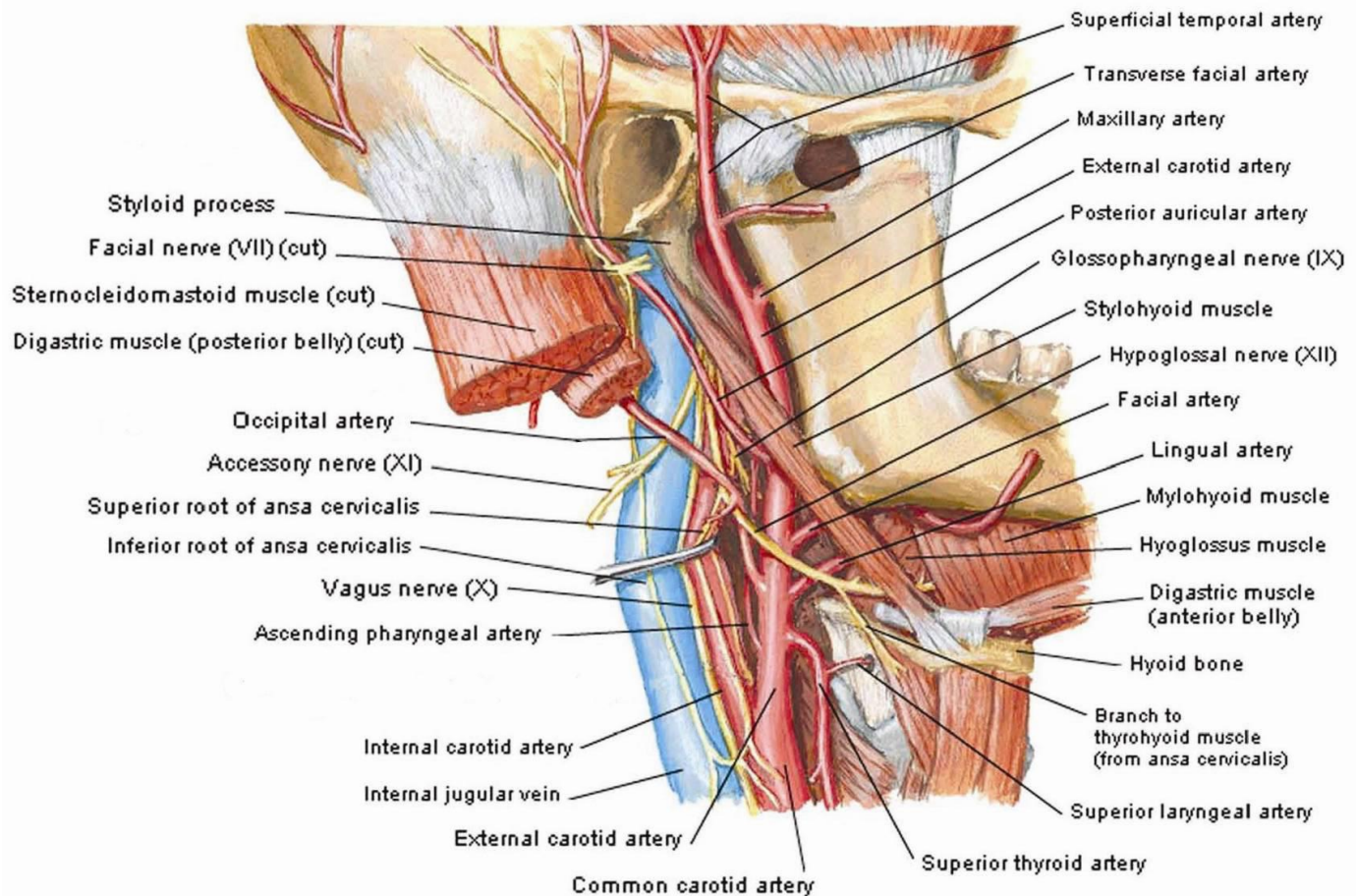
Deep Relations of E.C.A.

1. Parotid gland
2. Styloid process with
 - a. Styloglossus muscle
 - b. Stylopharyngeus muscle
3. Glossopharyngeal nerve
4. Pharyngeal branch of vagus
5. I.C.A.

N.B.: Structures 1, 2, 3, 4 present deep to E.C.A.,
superficial to I.C.A. or between E.C.A. and I.C.A.



Vessels and Nerves of Neck



Facial Artery

Origin:

It arises from the anterior surface of external carotid artery immediately above the lingual artery (above the greater horn of hyoid bone).

Parts and Course: It has 2 parts: cervical part in the neck and facial part in the face (**see head**).

Cervical part:

it ascends upwards on the side of pharynx (the middle and the superior constrictor muscles) where the superior constrictor muscle separates it from the tonsil. It then passes deep to the posterior belly of digastric muscle to enter the digastric triangle where it passes along the posterior border of submandibular gland grooving it.

It then hooks round the mandible to enter the face at the anteroinferior angle of masseter muscle to become the facial part in the face.

Branches of the cervical part:

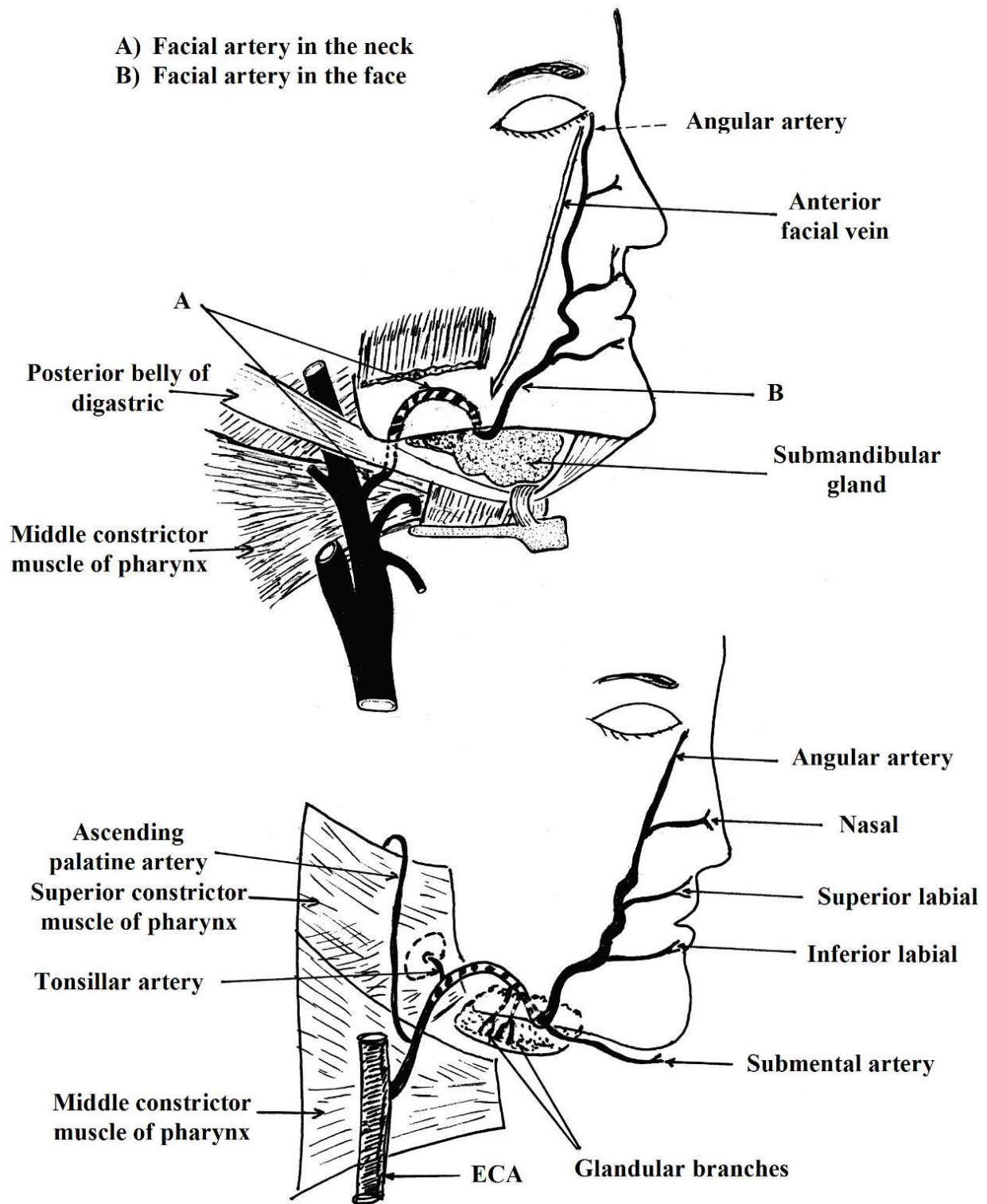
a- Tonsillar artery: The main artery of the tonsil, it pierces the superior constrictor muscle to reach and supply the tonsil.

b- Ascending palatine: It ascends on the pharynx (supplying it) to enter the gap above the superior constrictor muscle to reach and supplies the soft palate and palatine tonsil.

c- Glandular: To the submandibular gland and nearby muscles and skin.

d- Submental: The largest branch, passes along the lower border of the mandible to the submental triangle. It accompanies the nerve to mylohyoid and supplies the surrounding muscles. It anastomoses with sublingual, mylohyoid, inferior labial and mental arteries.

Facial Artery



Branches of Facial Artery

Lingual Artery

Origin: It arises from the external carotid artery opposite the tip of the greater horn of hyoid bone. It may arise from common stem with facial artery.

The course of lingual artery is divided into 3 parts by hyoglossus muscle.

A- First part: From external carotid artery till posterior border of hyoglossus muscle. It makes S-shaped loop lying on middle constrictor muscle

It gives suprahyoid artery which passes forwards on the lateral surface of hyoglossus

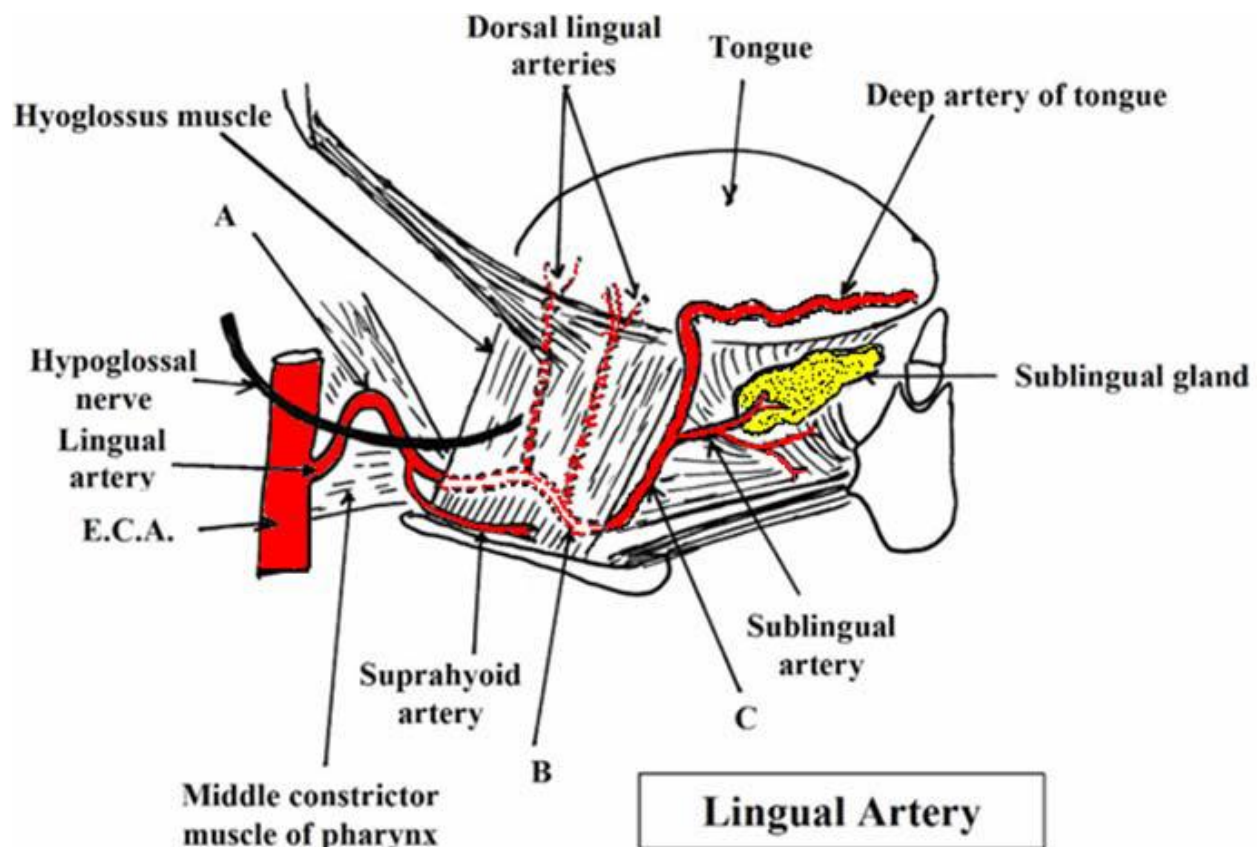
B- Second part: It passes deep to hyoglossus giving 2 dorsal lingual arteries which supply the posterior part of the tongue and end in the palatine tonsil supplying it.

C- Third part: Ascends along the anterior border of hyoglossus, lying on genioglossus,

It gives sublingual artery to sublingual gland then it ends as deep artery of tongue which supplies the anterior part of the tongue.

It is a tortuous artery

Sublingual artery: It arises from the third part of lingual artery, it supplies the sublingual gland, mylohyoid muscle, buccal and gingival mucosa. It gives branch which pierces the mylohyoid muscle to anastomose with the submental artery.



Occipital Artery

It arises from the posterior aspect of external carotid artery opposite the origin of facial artery. It passes upwards and backwards along the lower border of posterior belly of digastric then deep to mastoid process. It crosses the apex of posterior triangle to reach and supply the posterior part of the scalp.

Branches:

- **Two sternomastoid branches:** To sternomastoid muscle.
- **Mastoid branch:** Passes in the mastoid foramen to supply mastoid air cells.
- **Auricular branch:** Supplies the back of the auricle of the ear.
- **Muscular branches:** To digastric and stylohyoid muscles.
- **Stylomastoid artery (in 60% of cases):** It enters the stylomastoid foramen to supply the middle ear by posterior tympanic artery.

Posterior Auricular Artery

It arises from the posterior aspect of external carotid artery above the posterior belly of digastric muscle, runs along its upper border and then curves upwards behind the auricle to supply:

- **Auricular branches:** To posterior surface of the auricle of the ear.

- **Occipital branch:** To the scalp above and behind the auricle.
- **Stylomastoid artery (in 40% of cases):** It supplies the middle ear.

Ascending Pharyngeal Artery

Course:

It arises from the medial aspect of external carotid artery near its origin and ascends on the side of the pharynx deep to internal carotid artery to reach the base of the skull. It is the smallest branch of the external carotid artery.

Branches:

- **Pharyngeal branches:** To the pharynx, soft palate and tonsil.
- **Inferior tympanic artery:** Accompanies the tympanic branch of the glossopharyngeal nerve to supply the tympanic cavity.
- **Meningeal branches:** To the dura mater of posterior cranial fossa.

Internal Carotid Artery (I.C.A.)

Arises from CCA opposite the upper border of thyroid cartilage and ends at the base of brain by dividing into anterior and middle cerebral arteries.

At its beginning it shows a localized dilatation called the carotid sinus.

It has 4 parts: (Cervical-Petrous-Cavernous-Cerebral)

A- Cervical part: It lies in the neck, inside carotid sheath, medial to vagus and internal jugular vein.

Relations:

a- **Superficial relations:** External carotid artery with all structures lying superficial and deep to it.

b- **Deep and medial relations:** Superior laryngeal nerve, sympathetic chain, ascending pharyngeal artery and the wall of the pharynx.

c- **At the base of the skull:** Internal carotid artery lies anterior to the internal jugular vein with the lower 4 cranial nerves in between.

Branches: It has no branches in the neck.

B- Petrous part: In the carotid canal inside the petrous part, the artery runs upwards, then passes forwards and medially to leave it, and passes through the foramen lacerum on the upper surface of the cartilage that close it to enter the cavernous sinus.

Branches:

a- **Carotico-tympanic artery:** Enters tympanic cavity.

b- **Artery of pterygoid canal:** For the pharynx.

C- Cavernous part:

Inside the cavernous sinus, the artery runs in a sinuous course (upwards, then forwards and finally upwards). It leaves the sinus through the anterior part of its roof. The artery is related inferolaterally to abducent nerve (6th cranial nerve) and medially to the body of the sphenoid and sphenoid air sinus.

Branches: Inferior hypophyseal artery for the posterior lobe of the pituitary gland.

D- Cerebral part:

After leaving the roof of cavernous sinus, the artery passes backwards above the sinus for one cm, then it turns upwards to end just below the anterior perforated substance of the base of the brain by dividing into its two terminal branches.

Branches:

1- Superior hypophyseal arteries arise from the ICA after leaving the cavernous sinus, they supply the infundibulum of the pituitary gland.

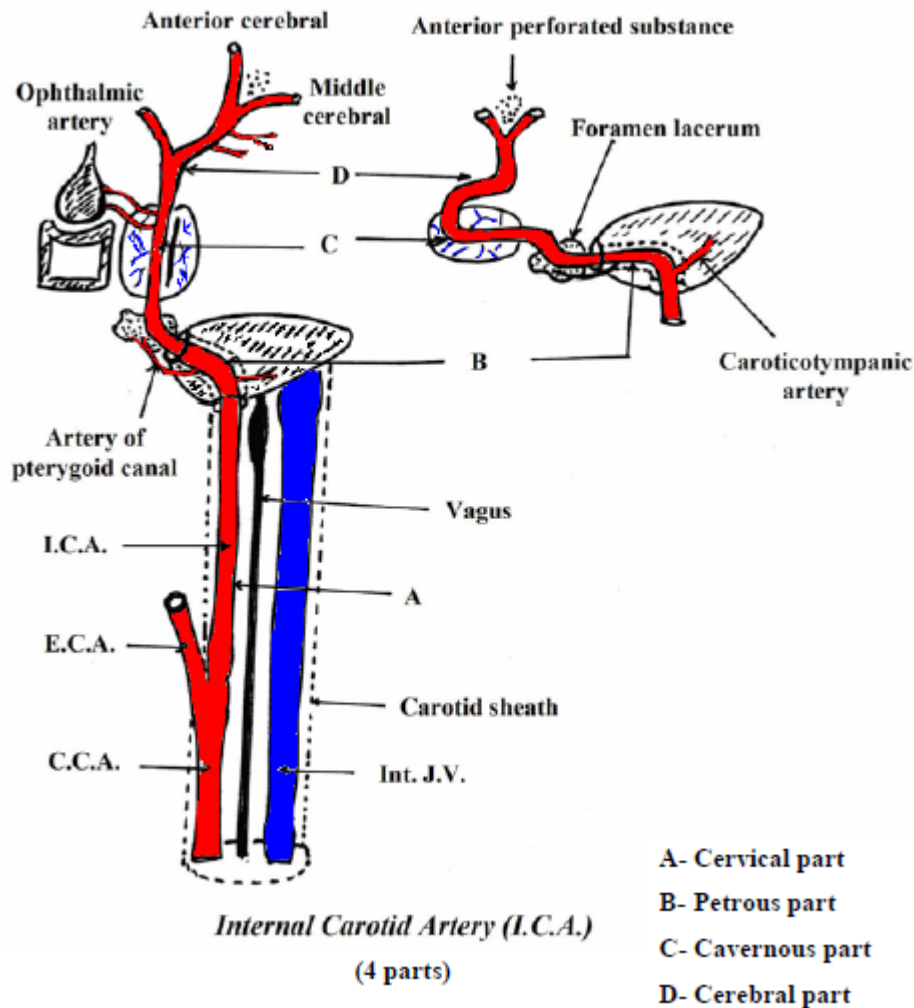
2- Ophthalmic artery.

3- Anterior cerebral artery.

4- Middle cerebral artery.

5- Posterior communicating artery. 6- Anterior choroidal artery.

N.B.: The petrous, cavernous and cerebral parts are called together the intracranial part of the internal carotid artery. It shows 6 bends to damp any sudden rise of pressure inside it.



Internal Jugular Vein (I.J.V.)

It **begins** at the jugular foramen as a continuation of the sigmoid sinus.

The right vein is usually larger than the left one.

It descends inside the carotid sheath lateral to the internal and common carotid arteries and vagus nerve.

It **ends** below by joining the subclavian vein to form the brachiocephalic vein behind the medial end of clavicle.

At its upper and lower ends, it forms the superior and inferior bulbs. The superior bulb is fixed to the margins of the jugular foramen.

Surface anatomy: A line is drawn from sternoclavicular joint till point between mastoid process and angle of mandible (It is the same surface marking of the CCA, ICA and carotid sheath).

Tributaries: From above downwards:

- 1- Inferior petrosal sinus. 2- Pharyngeal veins. 3- Common facial vein.
- 4- Lingual vein. 5- Superior thyroid vein. 6- Middle thyroid vein.

Relations:

Superficially:

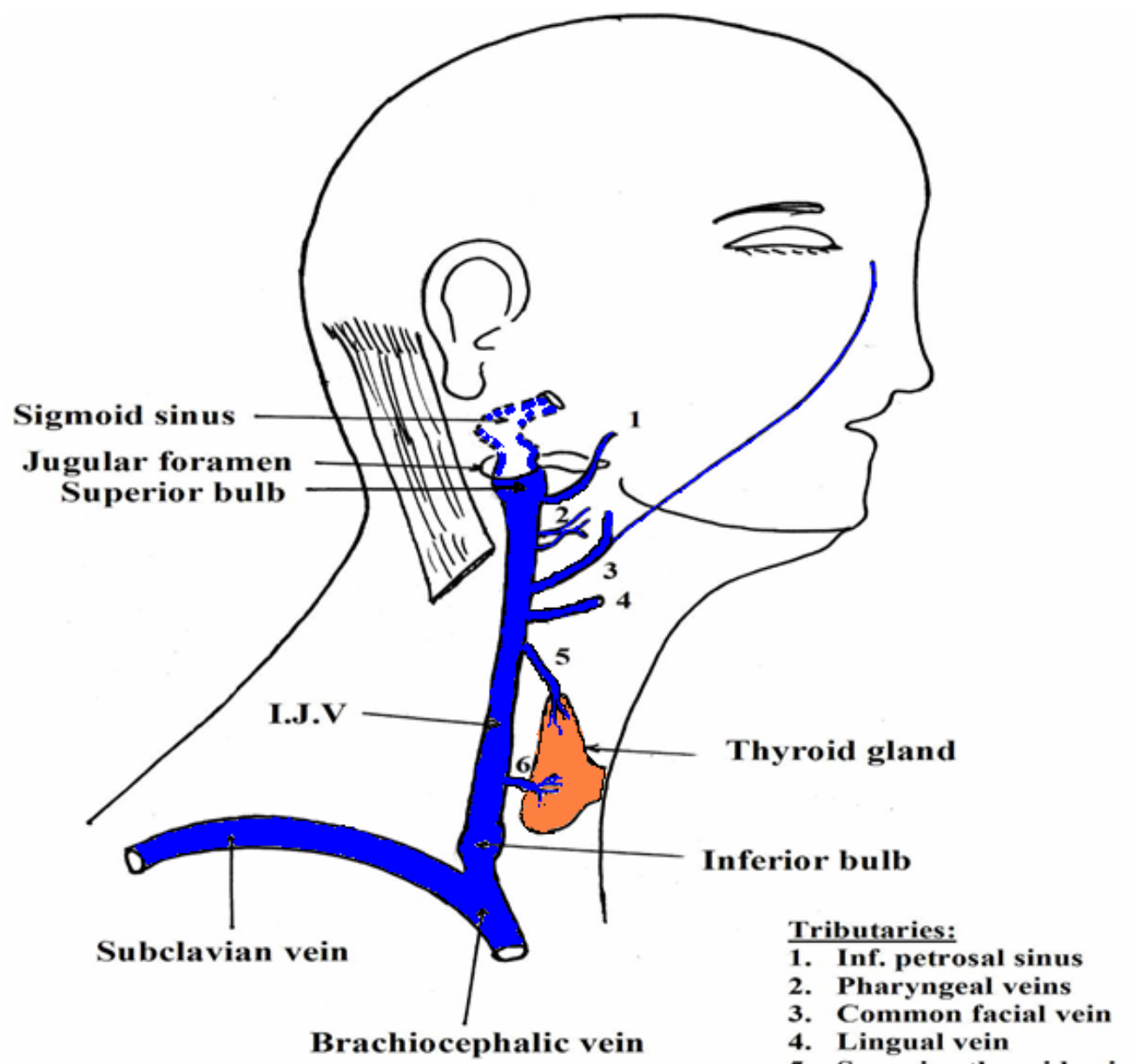
- 1- Styloid apparatus.
- 2- Posteromedial surface of parotid gland.
- 3- Posterior belly of digastric muscle with 2 small arteries related to it (occipital and posterior auricular).
- 4- Spinal part of accessory nerve crosses its upper part.
- 5- Inferior belly of omohyoid muscle.
- 6- Anterior jugular vein crosses its lower part.
- 7- Deep cervical lymph nodes.
- 8- Sternomastoid muscle.

Deep relations:

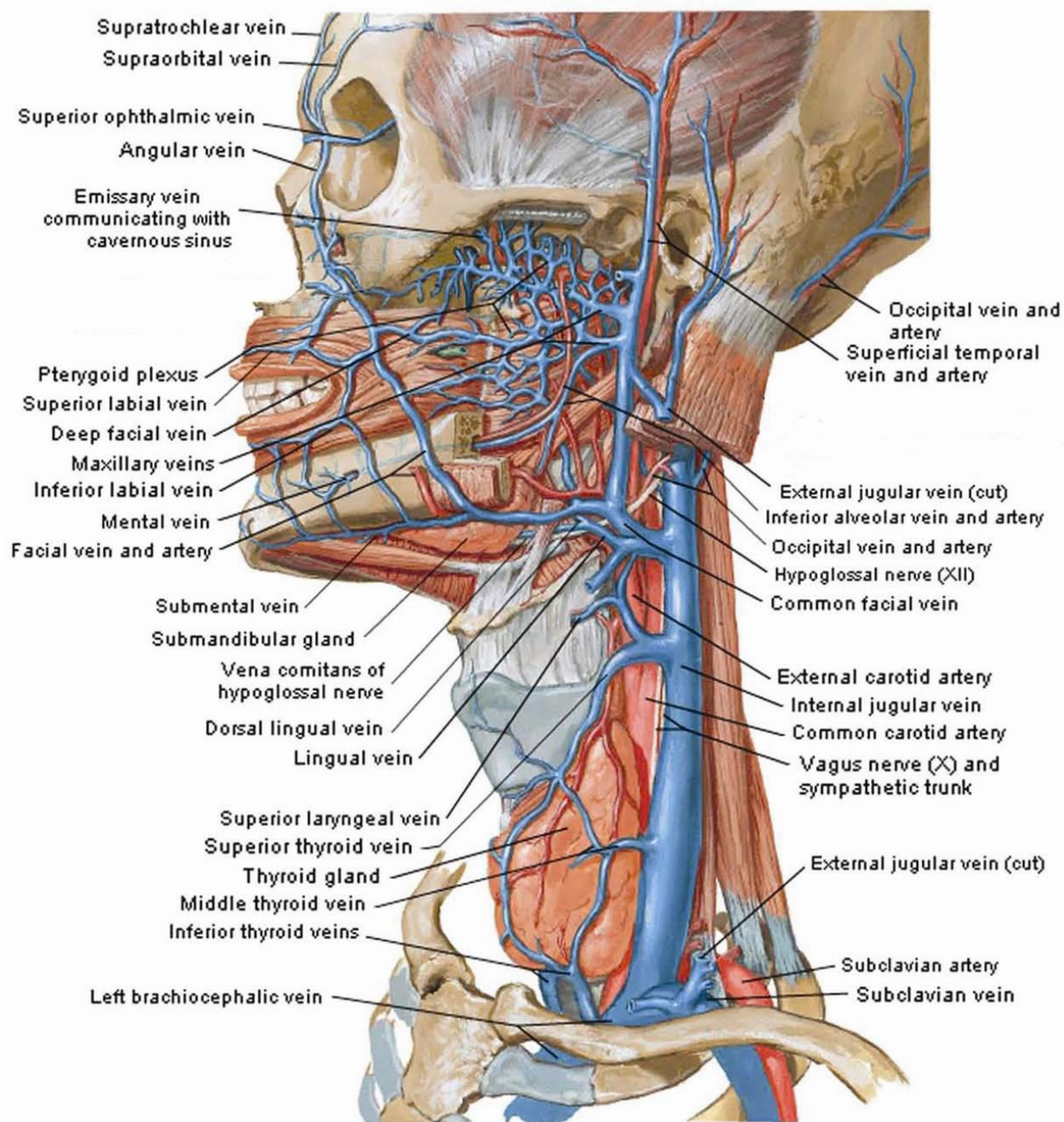
- 1- Transverse process of atlas vertebra. 2- Levator scapulae.
- 3- Scalenus anterior muscle. 4- Phrenic nerve.
- 5- First part of subclavian artery. 6- Thoracic duct (on the left side only).

Antero-medially:

- 1- **Below:** Common carotid artery and vagus nerve.
- 2- **Above:** Internal carotid artery and lower 4 cranial nerves.



Veins of Head and Neck



SUBMANDIBULAR REGION

Digastric and Submental Triangles

CONTENTS OF SUBMANDIBULAR REGION:

I. **Muscles:** Suprahyoid muscles, extrinsic muscles of tongue except palatoglossus and styloid muscles except stylopharyngeus.

II. **Nerves:** Lingual, hypoglossal and nerve to mylohyoid.

III. **Blood vessels:** Lingual and facial blood vessels.

IV. **Glands:** Submandibular and sublingual salivary glands.

V. **Ligaments:** Stylohyoid ligament.

VI. **Ganglion:** Submandibular ganglion.

Suprahyoid Muscles (4 muscles: Geniohyoid, Digastric, Mylohyoid & Stylohyoid).

1- Geniohyoid muscle:

Origin: Lower genial tubercle of mandible

Insertion: Anterior surface of the body of hyoid bone

Nerve supply: C1 via hypoglossal nerve

Action: elevation of hyoid bone

2- Digastric muscle: It consists of 2 bellies connected together by an intermediate tendon which is attached to hyoid bone by a strong fibrous loop.

Origin:

Anterior belly: from digastric fossa of mandible.

Posterior belly: from digastric fossa (mastoid notch) on the medial surface of mastoid process.

Insertion: Into intermediate tendon which is connected to hyoid bone by a fibrous loop.

Nerve supply:

Anterior belly: Nerve to mylohyoid (from mandibular nerve).

Posterior belly: Facial nerve.

Action:

a- Depression of the mandible, when the hyoid bone is fixed.

b- Elevation of hyoid bone in swallowing, when the mandible is fixed.

Relations of digastric muscle:

A- Relations of anterior belly:

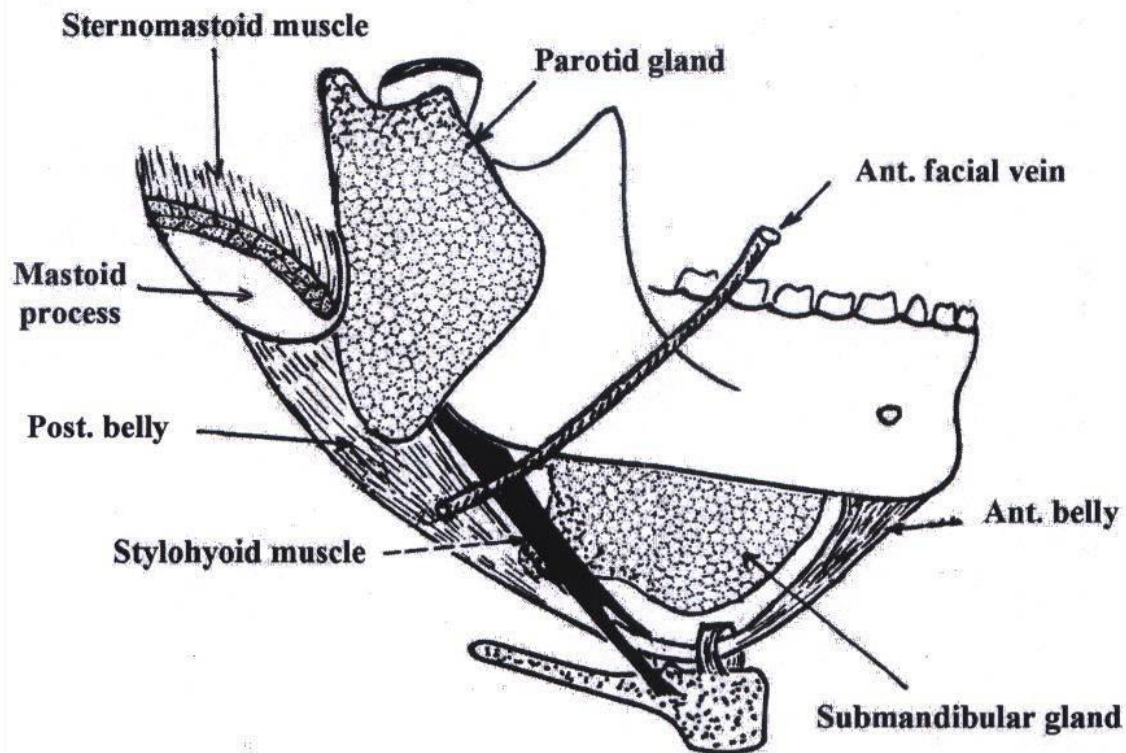
- **Superficial:** Skin and fascia.
- **Deep:** Mylohyoid muscle.

B- Relations of posterior belly:

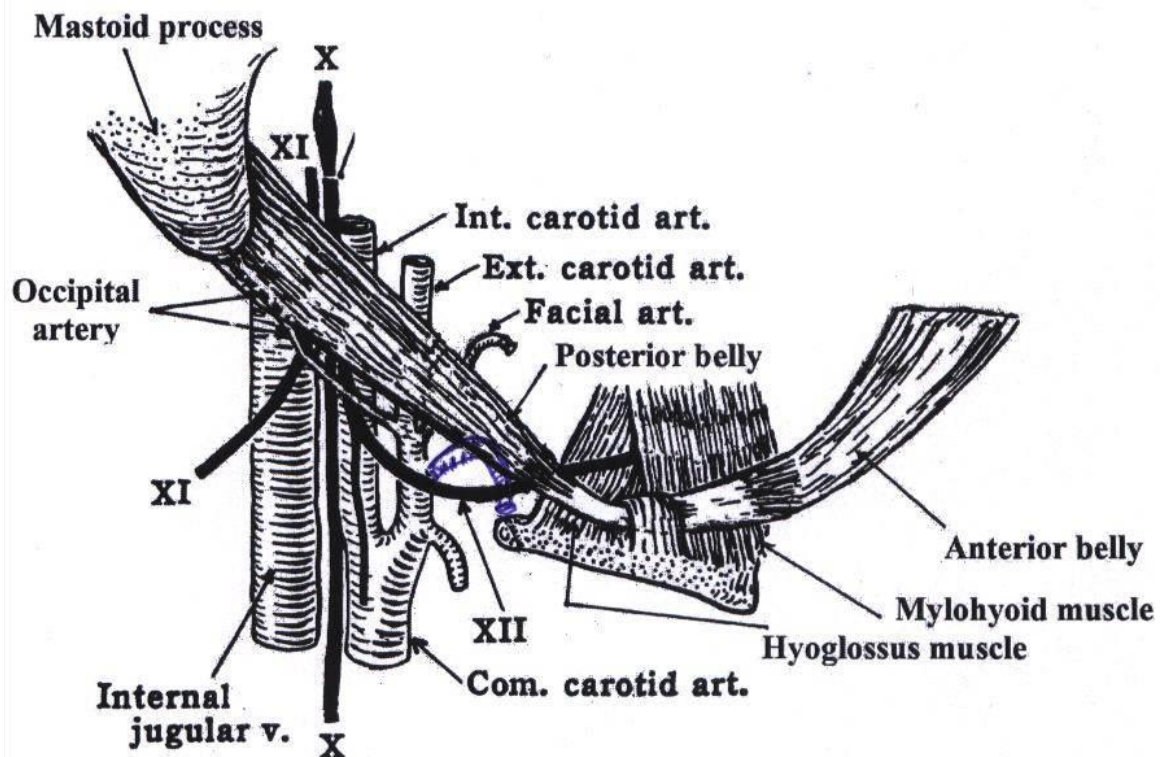
Superficial relations	Deep relations
1 vein Anterior or common facial vein	1 muscle: Hyoglossus
2 bones ▪ Mastoid process ▪ Angle of mandible	3 small arteries: Lingual, facial artery and occipital artery
2 glands ▪ Lower end of parotid gland ▪ Submandibular gland	3 large blood vessels: External CA, internal CA and internal jugular vein
3 muscles; sternomastoid, masseter, medial pterygoid	3 last cranial nerves: 10 th , 11 th and 12 th

C- Relations of borders:

- 1- The upper border is related to posterior auricular artery and stylohyoid muscle.
- 2- The lower border is related to occipital artery which passes backwards and to hypoglossal nerve which passes forwards.



Superficial Relations of the Posterior Belly of Digastric Muscle



3- Mylohyoid muscle:

It forms the great part of the floor of the mouth, so it is called "diaphragma oris".

Origin: The whole length of mylohyoid line of the mandible

Insertion:

a- Anterior and middle fibers inserted into the mylohyoid raphe.

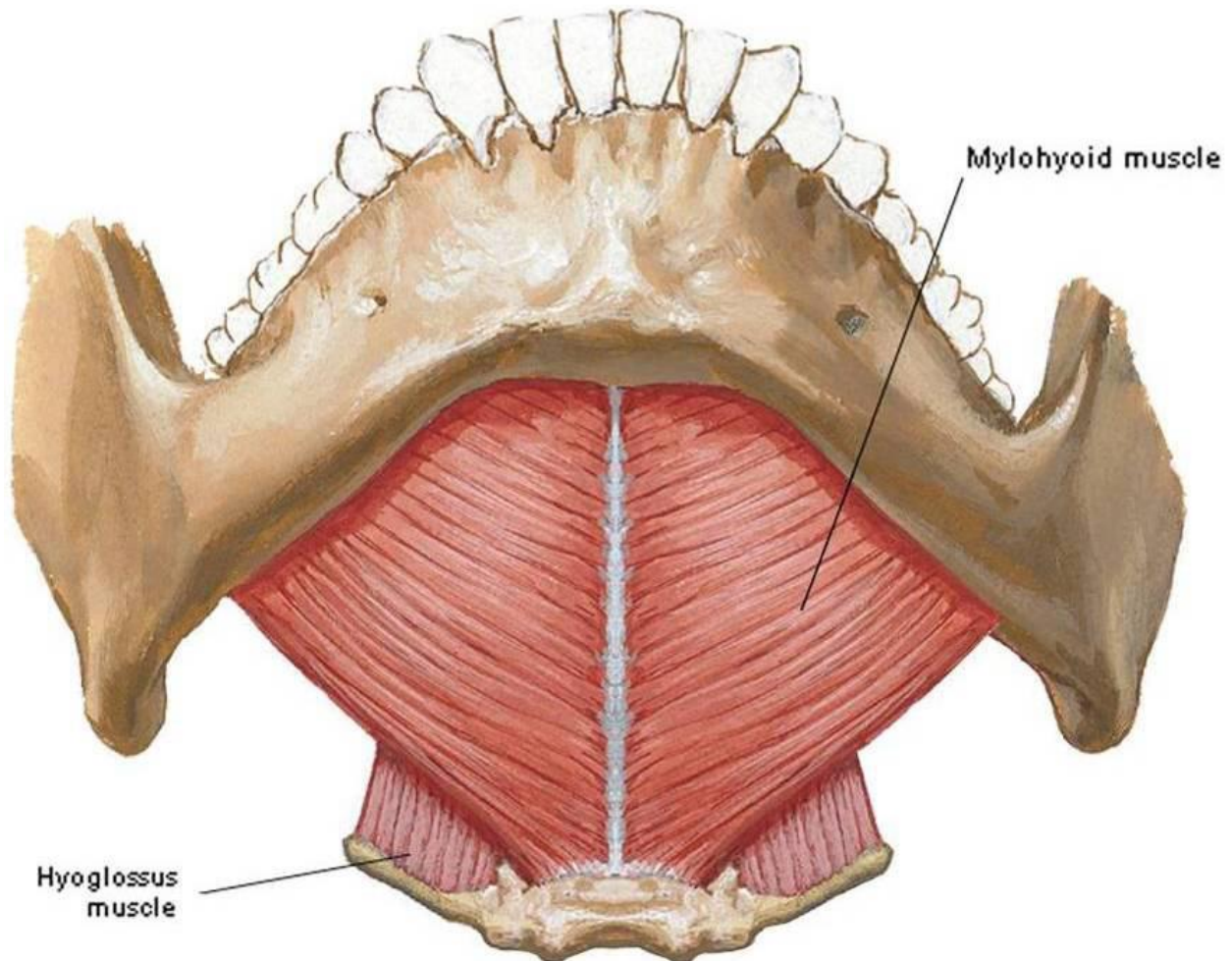
b- Posterior fibers inserted into body of hyoid bone.

Nerve supply: nerve to mylohyoid from inferior alveolar nerve (mandibular nerve).

Action:

a- Elevation of hyoid bone and floor of mouth.

b- Depression of mandible when hyoid bone is fixed.



Relations of mylohyoid muscle:

Superficial (lower) relations	Deep (upper) relation.
1. Skin and superficial fascia containing platysma & deep fascia 2. Anterior belly of digastric muscle 3. Superficial part of submandibular gland 4. Facial artery and its submental br. 5. Nerve to mylohyoid 6. Submental lymph nodes	1. Hyoglossus muscle. 2. <u>Structures anterior to hyoglossus:</u> Geniohyoid, genioglossus, sublingual gland with its vessels and 3rd part of lingual artery 3. <u>Structures lateral to hyoglossus:</u> Styloglossus muscle, lingual nerve and submandibular ganglion suspended to it, deep part of submandibular gland and its duct, hypoglossal nerve, vena comitants of hypoglossal nerve and suprahyoid artery.

4 muscles
2 glands
2 nerves
1 ganglion
1 artery

4- **Stylohyoid muscle** (see styloid apparatus).

Styloid Apparatus

It consists of styloid process and structures attached to it (3 muscles and 2 ligaments).

Muscle	Origin	Insertion	Nerve supply	Action
<i>Styloglossus</i>	Tip of styloid process	Whole side of the tongue	Hypoglossal nerve	Retraction and elevation of the tongue
<i>Stylohyoid</i>	Posterior surface of styloid process near the base	Junction of the body of hyoid bone with its greater horn by 2 slips	Facial nerve	Elevation of hyoid bone in swallowing
<i>Stylo-pharyngeus</i>	Medial surface of base of styloid process	Posterior border of thyroid cartilage	Glossopharyngeal nerve	Elevation of thyroid cartilage and shortening of pharynx in swallowing

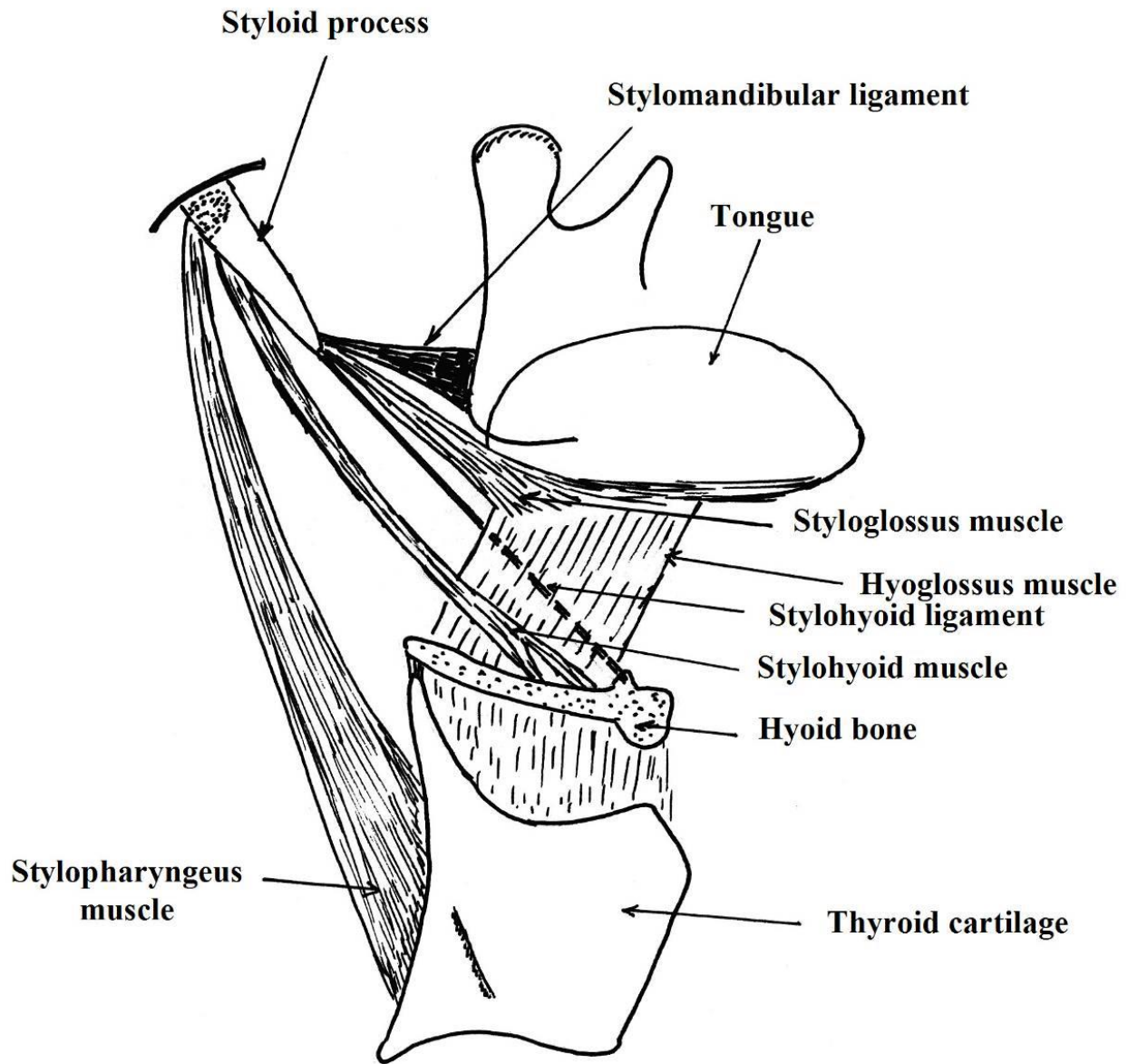
Styloid ligaments:

1- Stylohyoid ligament:

It extends from the tip of the styloid process till the lesser horn of hyoid bone.
It passes deep to hyoglossus.

2- Stylomandibular ligament:

It extends from the adjacent anterior surface of styloid process near its tip to the angle and posterior border of mandible. It is a thickened band of the general investing fascia of the neck. It separates 2 salivary glands; parotid and submandibular.



Styloid Apparatus

Extrinsic Muscles of the Tongue (4 muscles)

1- **Styloglossus:** See styloid apparatus.

2- **Palatoglossus:** See soft palate.

3- **Genioglossus:**

Origin: Upper genial tubercle of mandible.

Insertion:

a- **Upper and middle fibers** are inserted into the whole length of the ventral surface of the tongue.

b- **Lower fibers** are inserted into the anterior surface of the upper part of the body of hyoid bone near the midline.

Nerve supply: Hypoglossal nerve (12th cranial nerve).

Action:

1- **One muscle:** Protrudes the tongue to the opposite side.

2- **Both muscles:** Protrude the tongue forwards and depress the central part of tongue. They also prevent the tongue from falling backwards during sleep as this may obstruct the air pathway.

4- Hyoglossus:**Origin:**

The whole length of the upper surface of the greater horn of hyoid bone and front of body of hyoid.

Insertion:

Posterior 1/2 of the side of the tongue deep to styloglossus muscle.

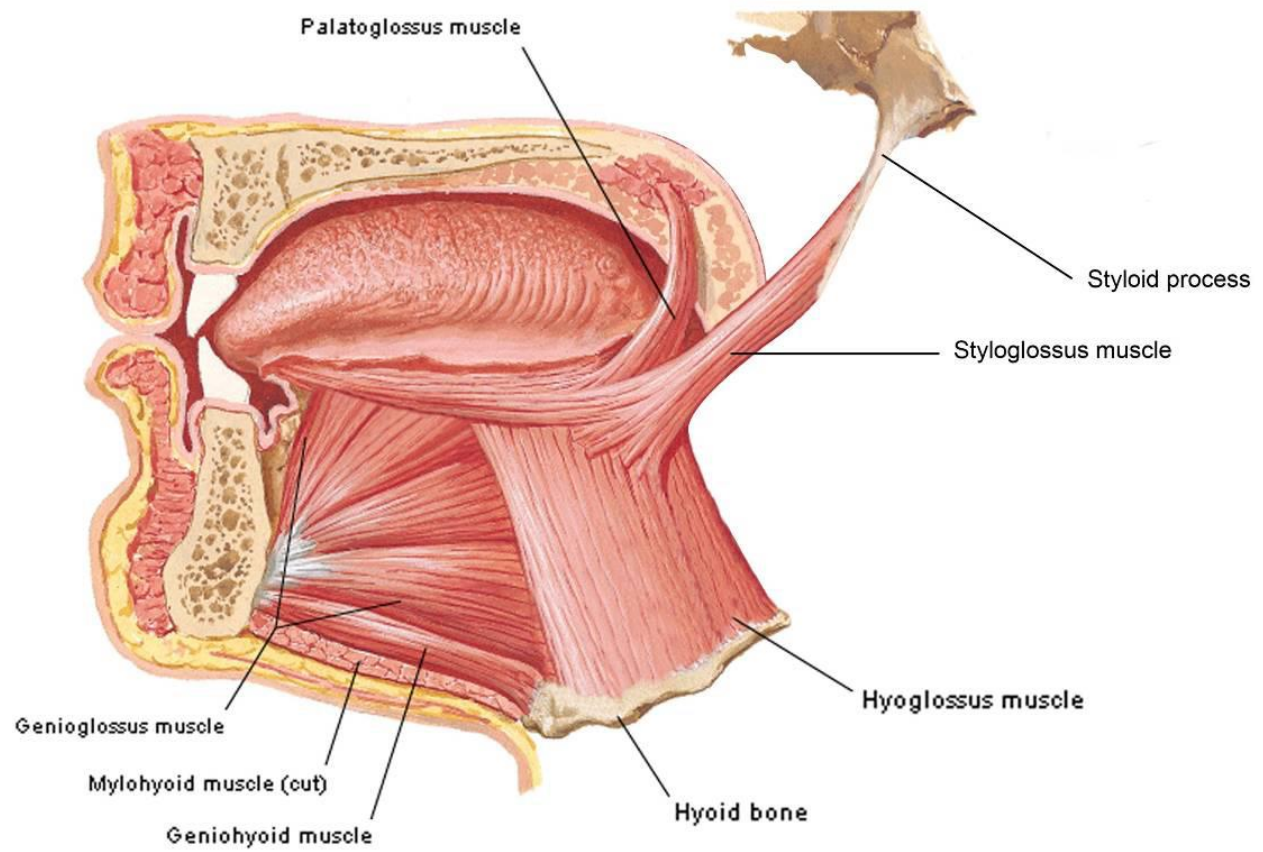
Nerve supply:

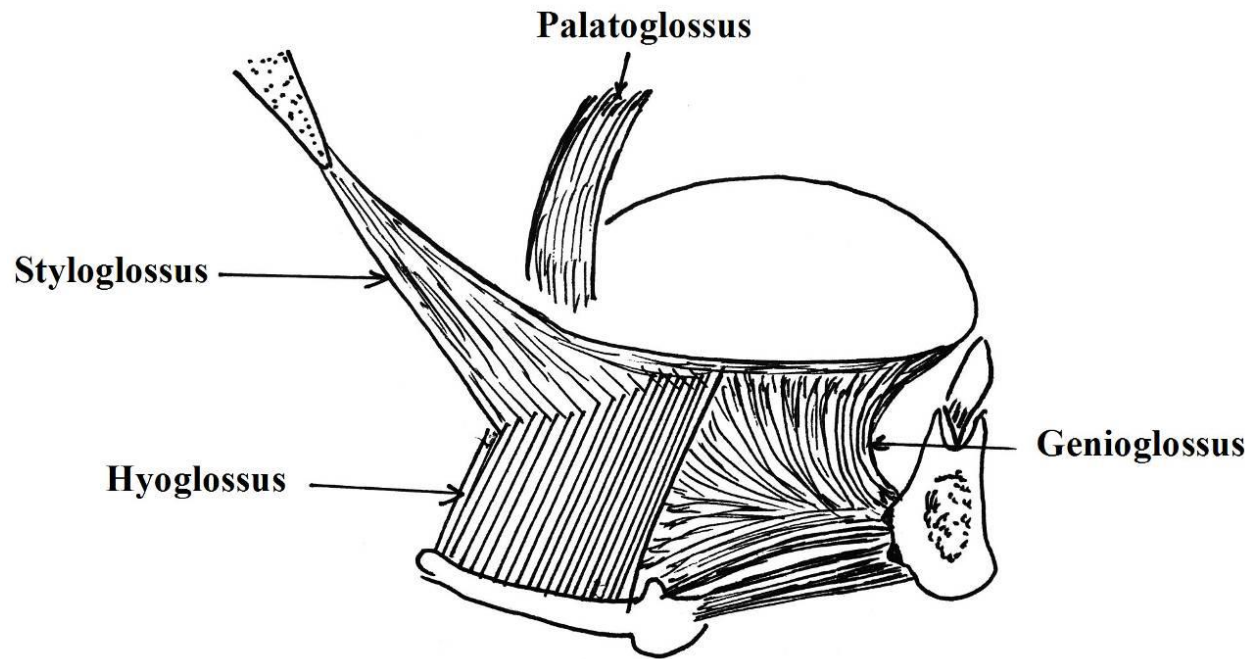
Hypoglossal nerve (12th cranial nerve).

Action:

Depression of the posterior ½ of the tongue against the floor of the mouth in sucking.

Muscles of Tongue





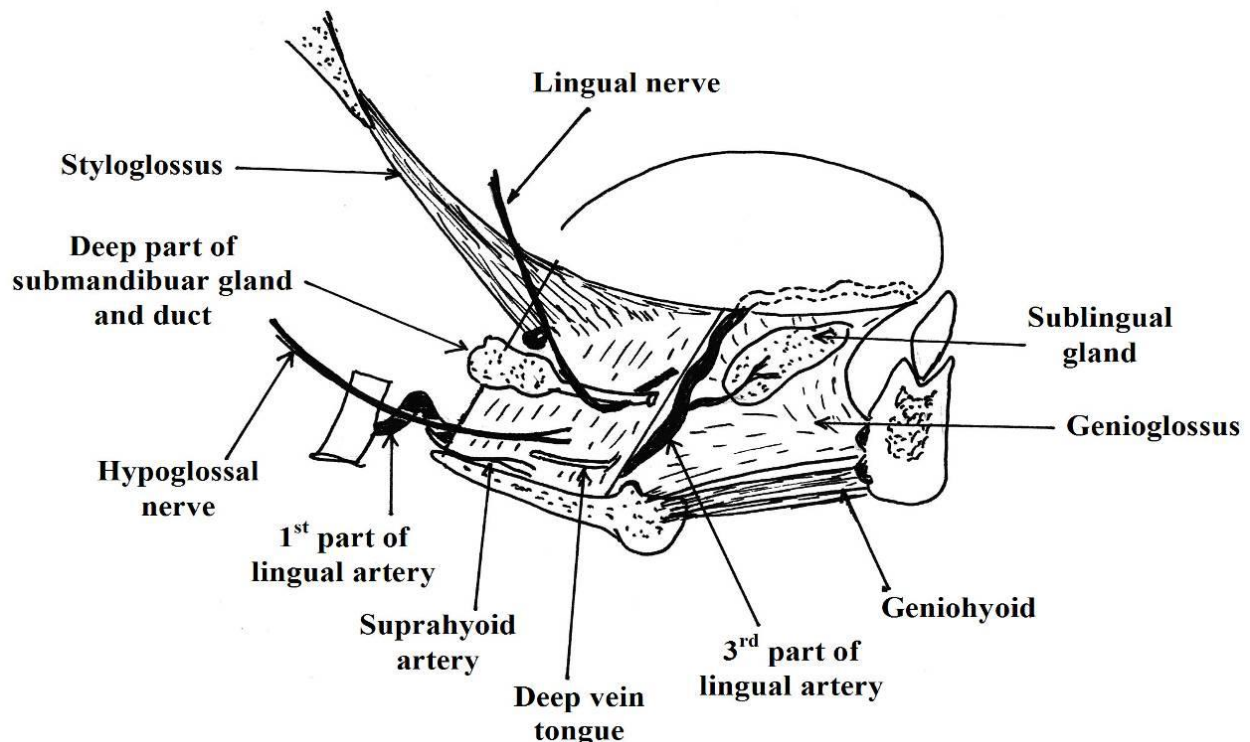
Extrinsic Muscles of tongue

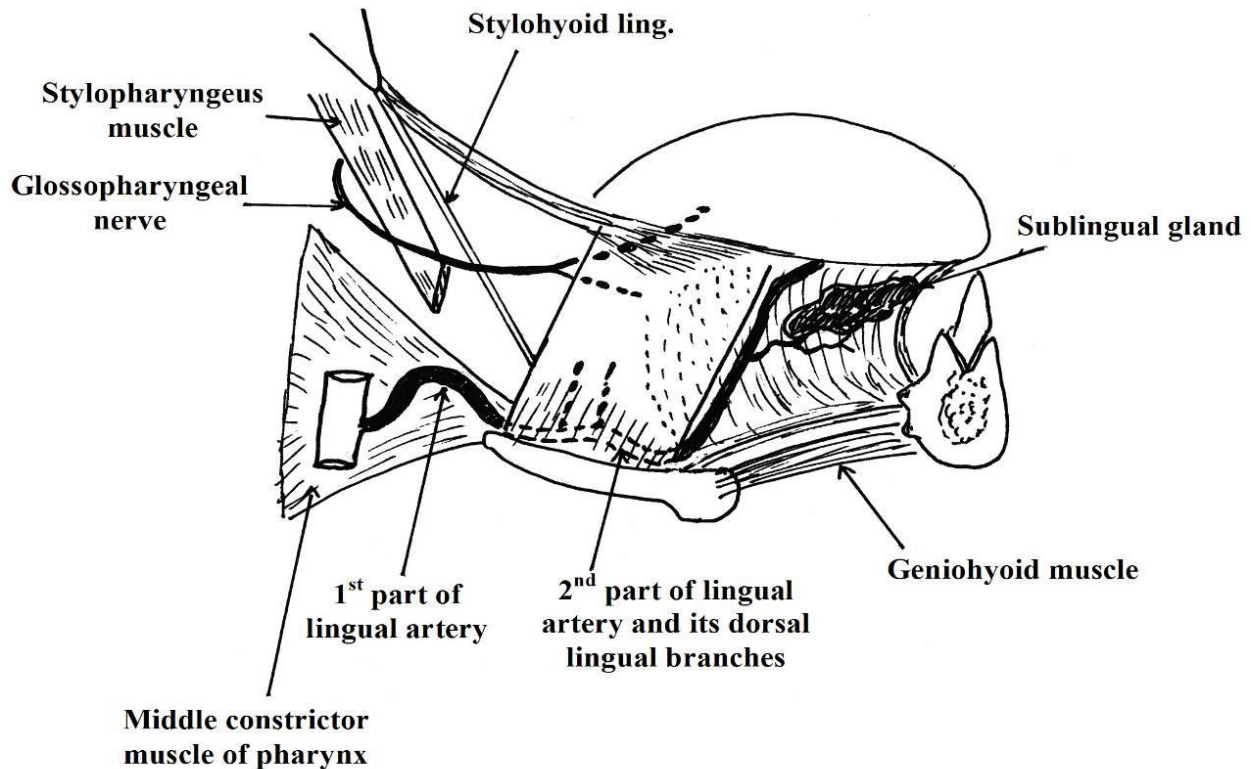
Relations of hyoglossus muscle:

Superficial (lateral) relations	Anterior relations (along anterior border)
• Styloglossus muscle • Lingual nerve hooking around submandibular duct • Deep part and duct of submandibular gland • Hypoglossal nerve with its vena comitans (Deep vein of the tongue) • Suprahyoid artery	- Geniohyoid muscle - Genioglossus muscle 3rd part of lingual artery - Sublingual gland and its blood vessels

All anterior and lateral relations are present deep to mylohyoid muscle

Posterior relations (along posterior border)	Deep (medial) relations
1st part of lingual artery - Middle constrictor muscle of pharynx - Glossopharyngeal nerve - Stylohyoid ligament and stylopharyngeus muscle	• 2nd part of lingual artery with its 2 dorsal lingual branches • Stylohyoid ligament • Glossopharyngeal nerve • Genioglossus-middle constrictor muscle of pharynx





Relations of Hyoglossus muscle

Submandibular Salivary Gland

It consists of 2 parts: Superficial and deep.

Superficial Part of Submandibular Gland

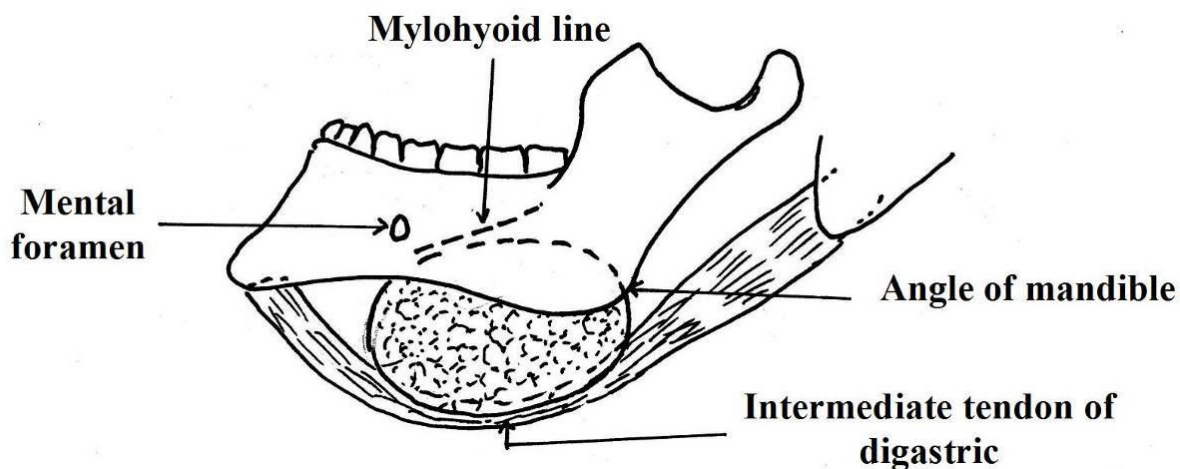
The main part of the gland which lies below (superficial) the mylohyoid muscle.

Position: It lies in the submandibular fossa of the mandible, extending from mylohyoid line (**above**), to overlap intermediate tendon of digastric (**below**), and from mental foramen (**in front**) to angle of mandible (**behind**).

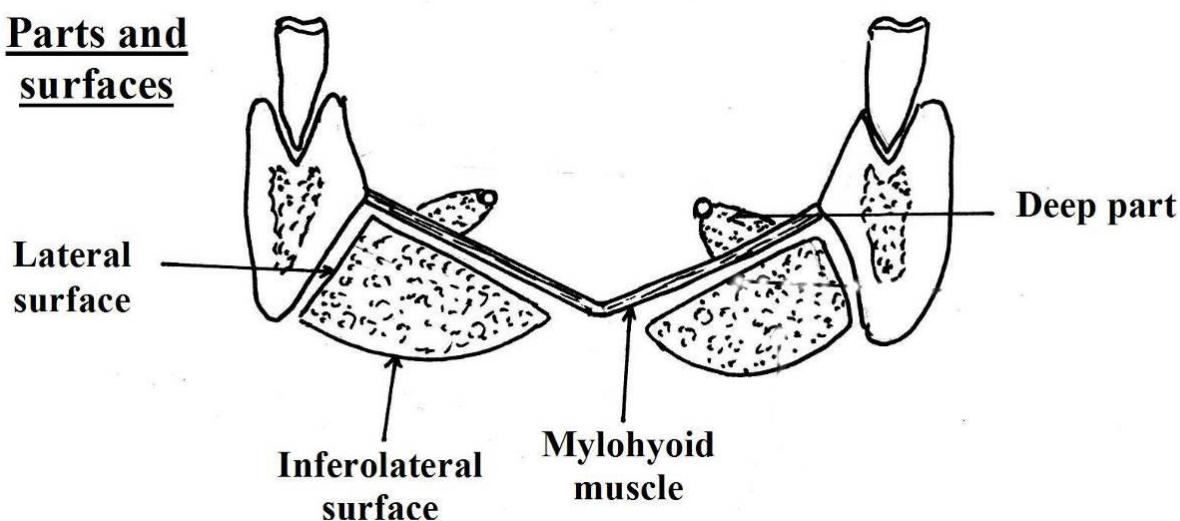
Surfaces and Relations: The gland is wedged between the body of mandible and mylohyoid muscle.

Superficial

Position



Parts and surfaces



The gland has 3 surfaces (lateral, medial, and inferolateral):

a- Lateral surface: It lies in contact with the inner surface of the body of mandible

and has the following relations:

- 1- Submandibular fossa of the mandible.
- 2- Insertion of medial pterygoid muscle.
- 3- Facial artery (between the gland and insertion of medial pterygoid).

b- Medial surface:

It has the following relations:

- 1- Mylohyoid muscle with nerve to mylohyoid and submental artery on the muscle (in the anterior part).
- 2- Hyoglossus muscle with all structures on its lateral surface (6 structures) (in the intermediate part).

- 3- Styloglossus muscle, stylohyoid ligament and glossopharyngeal nerve (in the posterior part).
- 4- The deep part of the gland and its duct project from this surface. These 2 structures run together between mylohyoid and hyoglossus muscles.

c- Inferolateral surface: It lies superficially below the mandible and has the following relations:

- 1- Submandibular lymph nodes.
- 2- Anterior facial vein.
- 3- Cervical branch of facial nerve.

All these structures are covered by skin, platysma and deep fascia.

Deep Part of Submandibular Gland

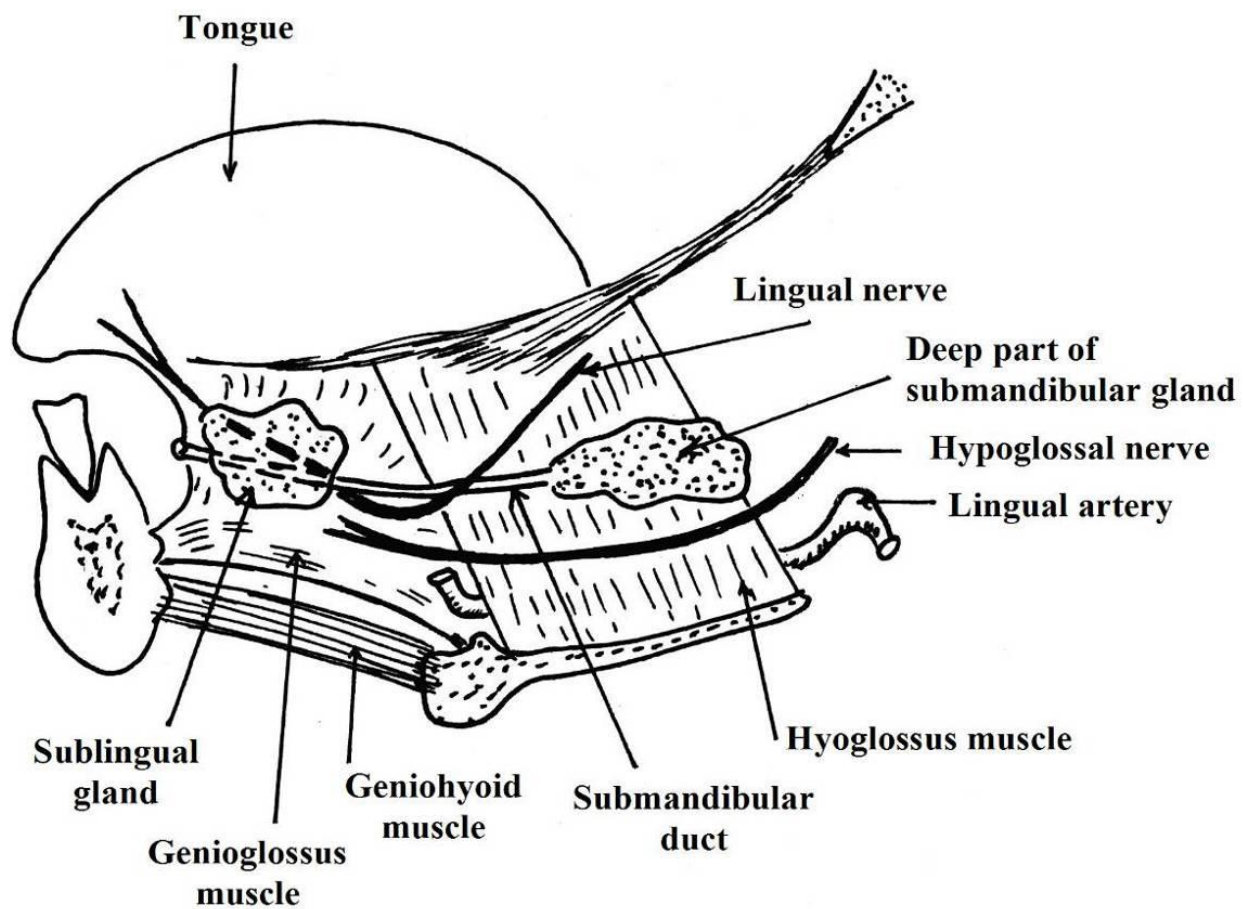
It is a thin flat tongue-like process that extends along submandibular duct, deep to mylohyoid, on the lateral surface of hyoglossus, with lingual nerve (above) and hypoglossal nerve (below).

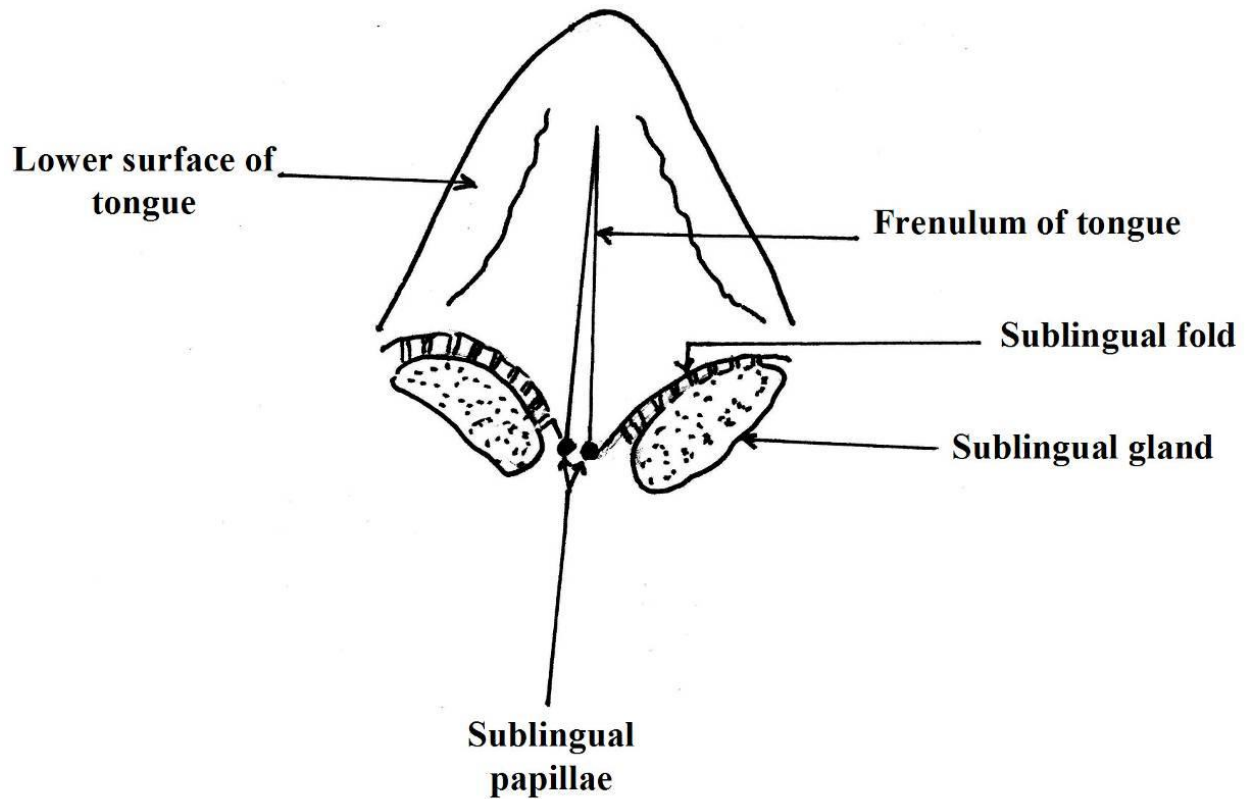
It is continuous with the superficial part of the gland around the posterior border of mylohyoid muscle.

Submandibular Duct:

It is 5 cm long.

It runs forwards through or along the deep part of the gland between the mylohyoid and hyoglossus muscles, with the lingual nerve (above) and hypoglossal nerve (below). At the anterior border of hyoglossus, the lingual nerve hooks around the duct (it double crosses the duct). It then continues forwards deep to sublingual gland on genioglossus to open in the floor of mouth in the summit of the sublingual papilla on both sides of the frenulum of the tongue.





The Openings of the Ducts of Submandibular and Sublingual glands

Applied Anatomy:

Stone in the submandibular duct:

It is very common, 50 times more than that of the parotid duct because:

- 1- The secretion is thick and viscid.
- 2- The duct drains its secretion against gravity.
- 3- The duct lies on the floor of mouth where any food particle may enter it and form a nucleus for the stone.

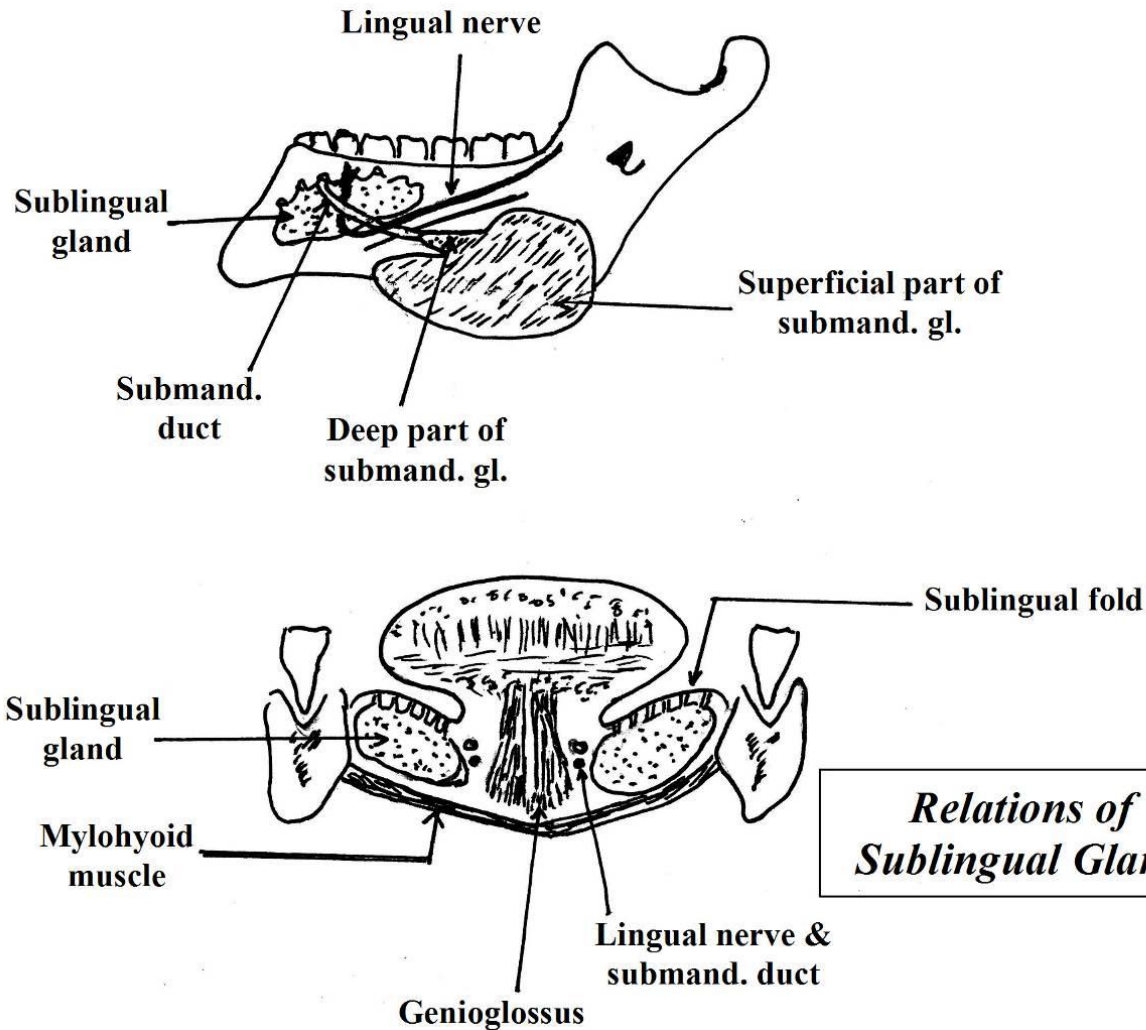
The patient complains of swelling in the submandibular region which increases in size during meals and returns to its normal size in between meals.

Sublingual Gland

It is almond shaped (3 cm long), which lies on genioglossus muscle and raises the mucous membrane of the floor of the mouth forming the sublingual fold. It has 8-12 fine ducts which open in the sublingual fold, but few of them open in the submandibular duct.

Relations:

- **Laterally:** Sublingual fossa of the mandible (above and in front of mylohyoid line).
- **Medially:** Genioglossus muscle with submandibular duct and lingual nerve between the gland and the muscle.
- **Below:** Anterior part of mylohyoid muscle.
- **Above:** Mucous membrane of mouth forming sublingual fold containing the orifices of the ducts of the gland.



Blood supply: Sublingual and submental blood vessels.

Nerve supply:

- 1- **Parasympathetic:** From chorda tympani nerve
- 2- **Sympathetic:** From plexus around lingual artery
- 3- **Sensory:** From lingual nerve.

Submandibular Ganglion

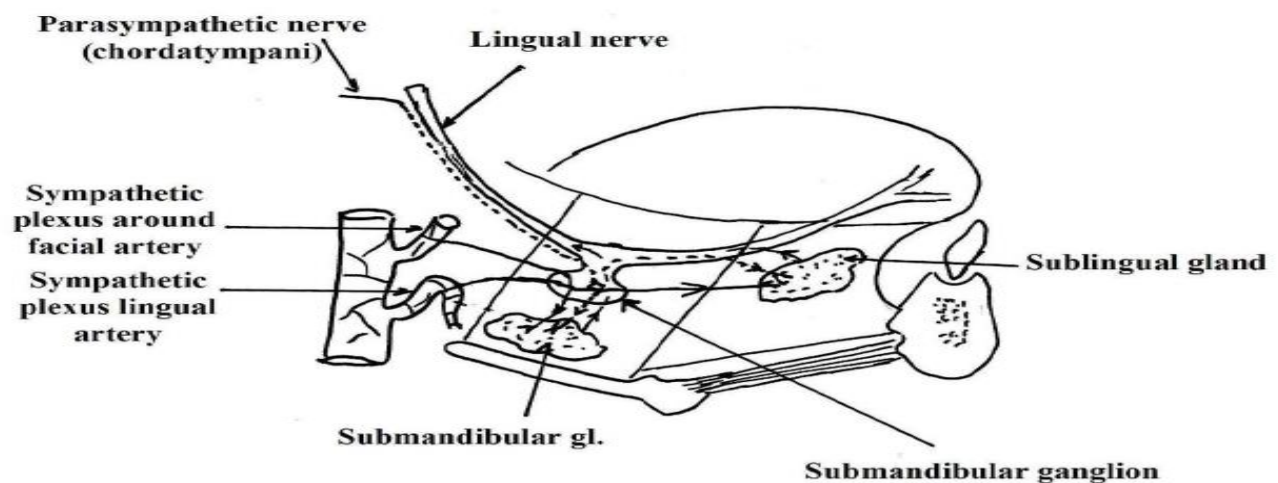
A small parasympathetic ganglion suspended from lingual nerve in the upper part of the lateral surface of hyoglossus muscle in the submandibular region.

Roots:

- 1- **Parasympathetic root:** Chorda tympani: it relays in the ganglion and its postganglionic fibers supply the submandibular and sublingual glands via the lingual nerve.
- 2- **Sympathetic root:** Sympathetic plexus around facial and lingual arteries: They traverse the ganglion without relay to reach and supply the submandibular and sublingual glands.
- 3- **Sensory root:** Short trunk connecting the lingual nerve to the ganglion. It contains sensory fibers which traverse the ganglion without relay to reach the submandibular and sublingual glands.

Branches:

Parasympathetic, sympathetic and sensory fibers to the submandibular and sublingual glands and to their blood vessels.



CRANIAL NERVES

Facial Nerve (7th cranial nerve)

Origin: It is a mixed nerve arising from 3 nuclei:

- 1- **Motor nucleus:** In pons.
- 2- **Parasympathetic nucleus:** Superior salivary nucleus in pons.
- 3- **Sensory nucleus:** Solitary nucleus in medulla.

Components of facial nerve: It is formed of 2 parts:

I. **Motor part:** Arises from the motor nucleus.

II. **Sensory and parasympathetic parts (Nervus intermedius):** It arises from the parasympathetic and sensory nuclei.

Course:

The facial nerve arises from pons, passes laterally to enter the internal auditory meatus, then enters the facial canal, which is directed laterally in the inner ear. It then passes backwards in the medial wall and downwards in the post wall of the middle ear and it finally leaves the facial canal via stylomastoid foramen.

Outside the skull: Facial nerve curves around the styloid process to enter the posteromedial surface of parotid gland, passes transversely superficial to the retromandibular vein and external carotid artery. It finally divides into 5 terminal branches (T-Z-B-M-C) (See nerve supply of the face).

Branches of Facial Nerve

(10 branches)

1- **Inside the petrous bone:** It gives 3 branches:

Nerve to stapedius, greater superficial petrosal and chorda tympani nerves.

2- **Just outside the skull as it emerges from stylomastoid foramen:** It gives 2 branches:

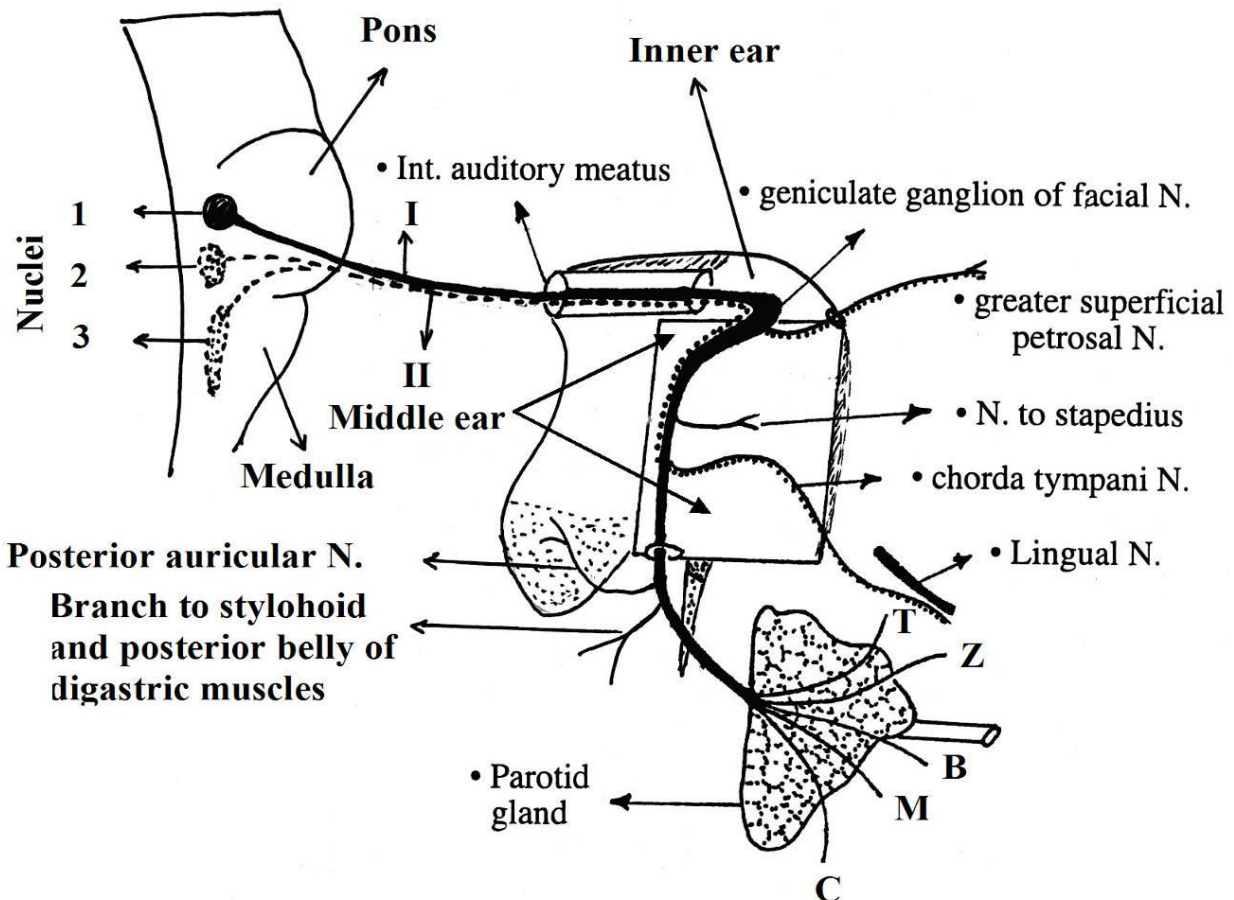
- a- Posterior auricular nerve supplies the occipital belly of occipitofrontalis muscle.
- b- Branch supplying to muscle stylohyoid and posterior belly of digastric muscles.

3- **In the face (inside parotid gland):** It gives 5 terminal branches:

Temporal (T) – Zygomatic (Z) – Buccal (B) – Mandibular (M) and Cervical (C):

they supply all muscles of face, scalp and platysma in the neck.

Facial Nerve (I & II)



Nervus Intermedius

It is the sensory and parasympathetic parts of facial nerve. It gives 2 branches:

A- Greater superficial petrosal nerve (see head).

B- Chorda tympani: It arises from facial nerve in the vertical part of facial canal, 6 mm above the stylomastoid foramen. It passes forwards through the substance of the tympanic membrane and leaves the middle ear through the petro-tympanic fissure (F). It enters the infratemporal fossa and joins the lingual nerve at acute angle deep to the lateral pterygoid muscle and distributes with its branches.

Branches:

1. Taste fibers to the anterior 2/3 of tongue.
2. Parasympathetic secretomotor fibers relay in the submandibular ganglion then the postganglionic fibers supply 2 glands: submandibular and sublingual glands.

Bell's palsy:

It is lower motor neurone lesion of the facial nerve which occurs spontaneously or without clear cause but recent MRI studies suggest that it may be caused by viral neuritis of the facial nerve at its beginning in the facial canal in the inner ear or in the adjacent brain stem.

It is characterized by:

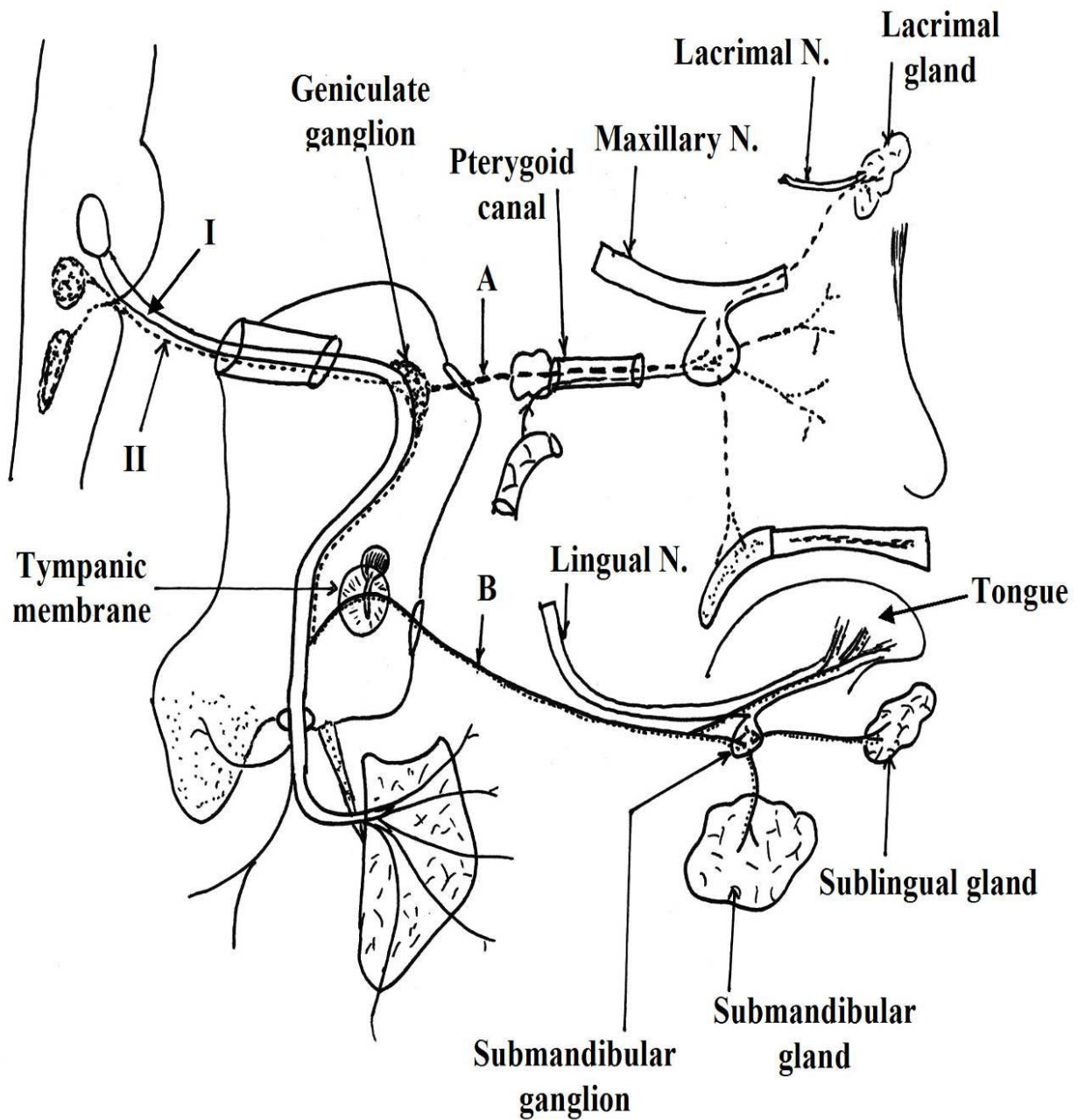
1- Paralysis of the facial muscles of expression of the affected side leading to:

- a. Dripling of saliva from the angle of the mouth in the affected side.
- b. Accumulation of food in the vestibule of mouth in the affected side.
- c. The patient can not close his eye in the affected side (the eyes remain open even during sleep).
- d. Deviation of the angle of the mouth to the opposite side (healthy side).

2- Decrease lacrimation in the eye of the affected side which is controlled by the neurons of greater superficial petrosal nerve.

3- Hyperacusis (decreased tolerance of loud noise in the ear of the affected side due to paralysis of stapedius muscle).

In the vast majority of cases of Bell's palsy spontaneous full recovery of the case take place after few weeks.



Branches of the Nervus Intermedius (II)

- A) Greater superficial petrosal
- B) Chorda tympani

Glossopharyngeal Nerve (9th cranial nerve)

Origin: From the medulla by 4 nuclei; motor (nucleus ambiguus), parasympathetic (inferior salivary), and two sensory nuclei: Taste (nucleus solitaries) and general sensation (spinal nucleus of trigeminal).

Course: It leaves the skull through jugular foramen, descends vertically in the upper part of carotid sheath between ICA and IJV deep to the styloid process, then it leaves the carotid sheath curving forwards with the stylopharyngeus muscle between ICA and ECA. The nerve enters the pharynx in the 2nd gap between superior and middle constrictor muscles of pharynx, then it passes deep to hyoglossus muscle and divides into its branches.

Branches:

1- Motor branch: For one muscle only (stylopharyngeus muscle).

2- Tympanic branch: It is a parasympathetic and sensory nerve. It enters the tympanic cavity, uniting with the caroticotympanic sympathetic nerves to form the tympanic plexus. It supplies the mucosa of middle ear, auditory tube, mastoid antrum and parotid gland via the lesser superficial petrosal nerve (see Parotid).

3- Sensory branches:

a- **Tonsillar branches:** For palatine tonsil and soft palate.

b- **Lingual branches:** For the general sensation and taste of posterior 1/3 of tongue. One of these branches supplies the vallate papillae and the mucosa near the sulcus terminalis.

c- **Pharyngeal branches:** They are 2-3 branches, uniting with the pharyngeal branches of vagus and superior cervical sympathetic ganglion to form the pharyngeal plexus on the surface of the middle constrictor muscle of pharynx.

The pharyngeal plexus supplies the mucosa of oropharynx.

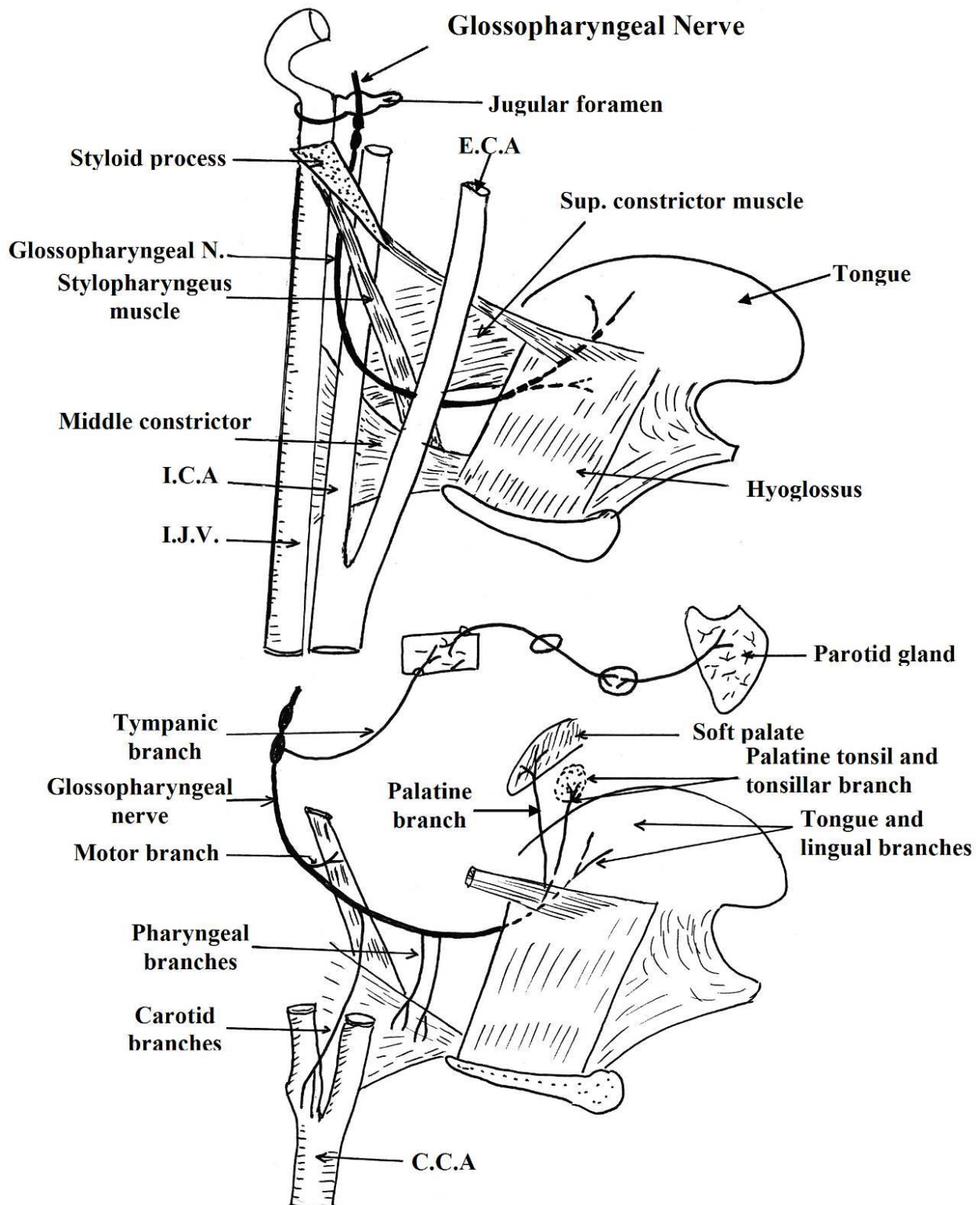
d- **Carotid branches:** Supply the carotid sinus and carotid body.

Applied anatomy:

Injury of glossopharyngeal nerve:

1- Absence of taste in the posterior 1/3 of tongue.

2- Absence of Gag reflex on the side of the lesion.



Branches of Glossopharyngeal Nerve

Vagus Nerve (10th cranial nerve)

Origin: From medulla oblongata by 4 nuclei:

- **Motor:** Nucleus ambiguus.
- **Parasympathetic:** Dorsal nucleus of vagus.
- **Sensory nucleus:** Nucleus solitarius.
- **Sensory nucleus:** Spinal nucleus of trigeminal.

Course: From the medulla, it passes through jugular foramen; descends in the neck inside the carotid sheath between IJV and ICA, then between IJV and CCA, crosses in front of the 1st part of subclavian artery to enter the thorax.

It has 2 ganglia; superior ganglion inside jugular foramen and inferior ganglion inside carotid sheath.

Branches:

1- Two branches from superior ganglion inside jugular foramen:

- **Meningeal branch** for the dura of posterior cranial fossa.
- **Auricular branch** for external auditory meatus and ear drum.

2- Two cardiac branches (superior and inferior):

They end in the cardiac plexuses to supply the heart.

3- Pharyngeal branch:

It consists mainly of filaments from the cranial root of accessory nerve.

It arises from inferior ganglion of vagus, passes between ECA and ICA to reach and take share in the pharyngeal plexus, on the middle constrictor muscle of pharynx.

It supplies all muscles of pharynx except stylopharyngeus and all muscles of the soft palate except tensor palati.

4- Laryngeal branches:

a- **Superior laryngeal nerve:** From the inferior ganglion of vagus, passes deep to both ECA and ICA and divides into two branches:

- **Internal laryngeal nerve (sensory):** It pierces thyrohyoid membrane; it supplies mucosa of larynx above the vocal cords and epiglottis.
- **External laryngeal nerve (motor):** It accompanies superior thyroid artery to reach and supply cricothyroid muscle. It gives a twig to inferior constrictor muscle of pharynx.

b- Recurrent laryngeal nerve (RLN):

The right nerve arises from right vagus in the root of neck, hooking around the subclavian artery, the left nerve arises from left vagus in the thorax hooking around the arch of aorta. Both nerves ascend between

trachea and oesophagus, to enter the larynx deep to the lower border of inferior constrictor muscle of pharynx. At the lower border of thyroid gland the RLN ascends within the branches of inferior thyroid artery.

Branches of RLN:

Motor: For all muscles of larynx except one muscle (cricothyroid). It gives a twig to inferior constrictor muscle of pharynx.

Sensory: For mucosa of larynx below the vocal cords.

Branches for trachea, oesophagus and heart.

Applied Anatomy:

Injury of vagus nerve leads to:

1- Tachycardia and cardiac arrhythmia.

2- Vasovagal attack: reflex of involuntary nervous system due to stimulation of vagus nerve causes: bradycardia, hypotension and fainting attack.

Causes: some persons when they seen **blood** or subjected to any type of **fear**.

3- Injury of the pharyngeal branch leads to dysphagia and deviation of the soft palate (uvula) to the opposite side.

4- Injury of laryngeal nerves:

A- Injury of the external laryngeal nerve: The close relation of the external laryngeal nerve to the superior thyroid artery, the nerve may be cut when the artery is ligated and clamped during thyroid gland lobectomy.

Injury of the external laryngeal nerve leads to paralysis of the cricothyroid muscle and relaxation of the vocal cords these leads to:

1. The vocal cords become flaccid and flappy allowing excess leakage of air.

2. The voice become weak, low pitched and breathy voice accompanied by mild hoarseness. These effects occur in bilateral lesions, but in unilateral lesion these effects may not be noticeable.

B- Injury of recurrent laryngeal nerves (RLN):

I. Unilateral RLN injury: It is common on the left side because it has long course in the thorax and the neck

Causes:

a. In the thorax: RLN is damage by either cardiac surgery, trauma or bronchogenic carcinoma which is the commonest cause.

b. In the neck: injury of Rt. or Lt. RLN in the neck can occur due to thyroid gland operation, cancer thyroid and penetrating neck trauma.

II. Bilateral RLN injury: These injuries can occur during thyroidectomy or by cancer thyroid gland.

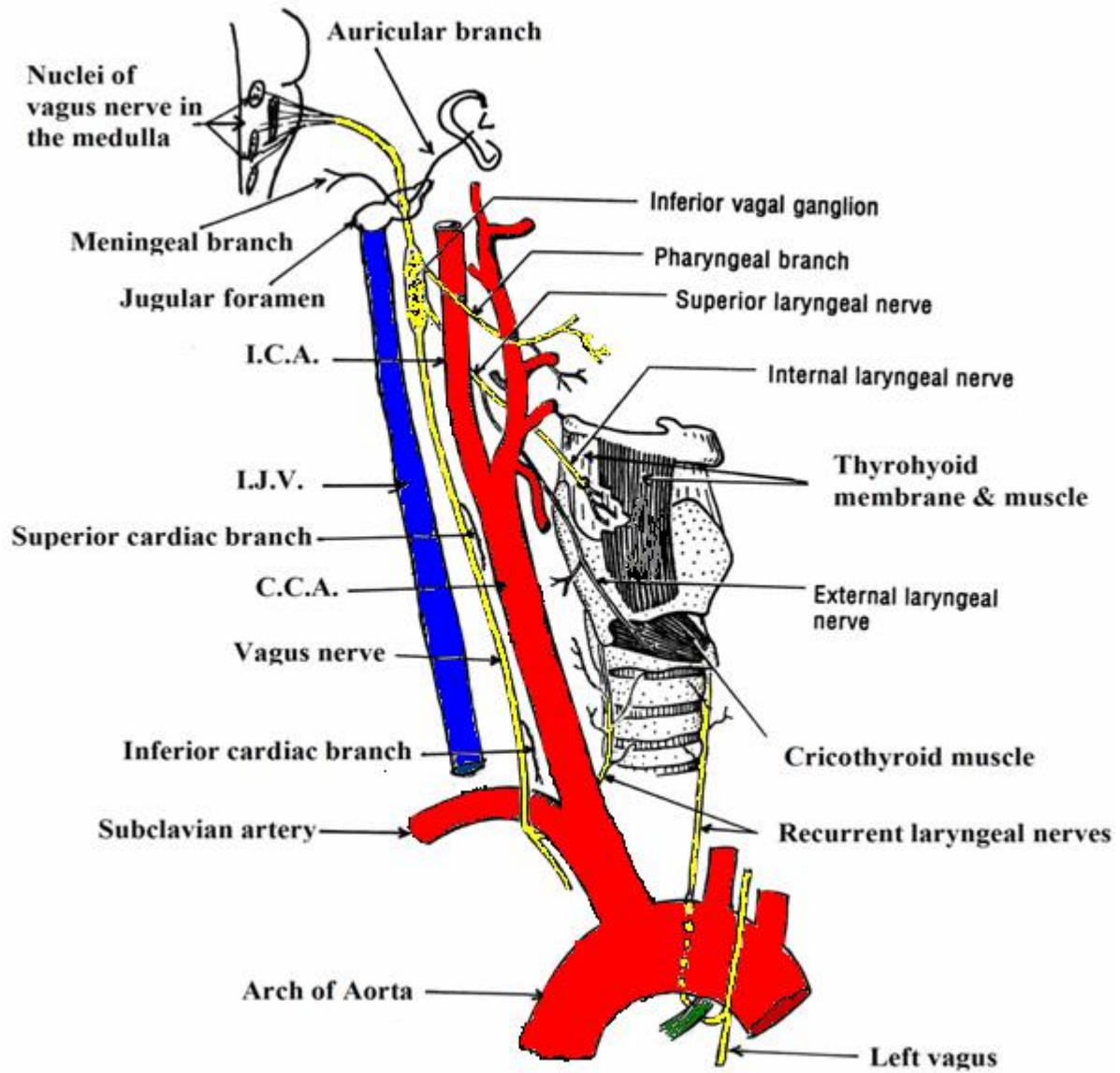
Clinical picture of the unilateral RLN injury

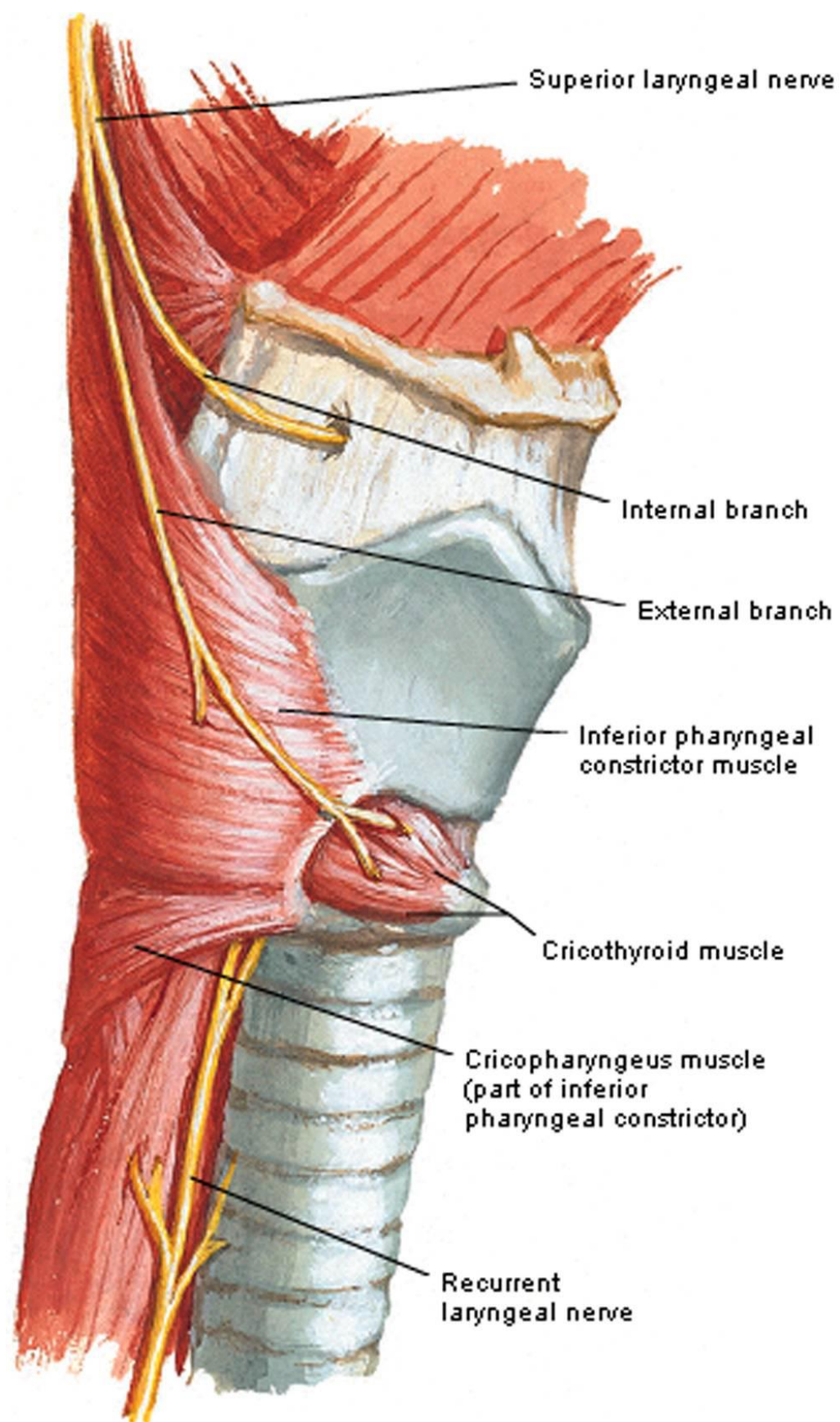
It leads to hoarse breathy voice, this hoarseness may be permanent or may become less severe with time as the opposite cord has the ability to hyperadduct and oppose the paralysed cord and thus close the glottis during phonation and coughing and these effects are enough to restore the full voice quality.

Clinical picture of the bilateral RLN injuries

1- Partial bilateral RLN injuries: induces paralysis of both abductors muscles of the vocal cords leading to adduction of the vocal cords leading to dyspnea and stridor. In this case tracheostomy is necessary for life.

2- Complete bilateral RLN injuries: induces paralysis of the abductor & adductor muscles of the vocal cords putting the vocal cords in position midway between abduction and adduction (cadaveric position) leading to aphonia = lose of speech.





Pharyngeal Plexus

A plexus of nerves that lies on the middle constrictor muscle of pharynx.

It consists of three parts (motor, sensory and sympathetic):

a- Motor part: Pharyngeal branch of vagus.

b- Sensory part: Pharyngeal branch of glossopharyngeal.

c- Sympathetic part: Pharyngeal branch of superior cervical sympathetic ganglion.

Branches:

a- Motor to all muscles of pharynx except stylopharyngeus (by glossopharyngeal nerve).

b- Motor to all muscles of palate except tensor palati (by nerve to medial pterygoid).

c- Sensory to mucosa of oropharynx.

Accessory Nerve (11th cranial nerve)

Two Roots:

A- Spinal root:

It arises from upper 5-6 segments of spinal cord. It ascends to enter the cranial cavity through foramen magnum, then it curves laterally, leaving the skull via jugular foramen to enter the neck. It descends between ICA and IJV deep to styloid process.

Then it passes backwards crosses IJV, deep to posterior belly of digastric muscle. It pierces the sternomastoid muscle at the junction of its upper and middle 1/3 to reach the posterior triangle of neck about the middle of the posterior border of the muscle.

It crosses the triangle within the deep fascia of its roof on the surface of the levator scapulae muscle and leaves the triangle deep to trapezius muscle, about 5 cm above the clavicle. It supplies 2 muscles (sternomastoid and trapezius).

B- Cranial root:

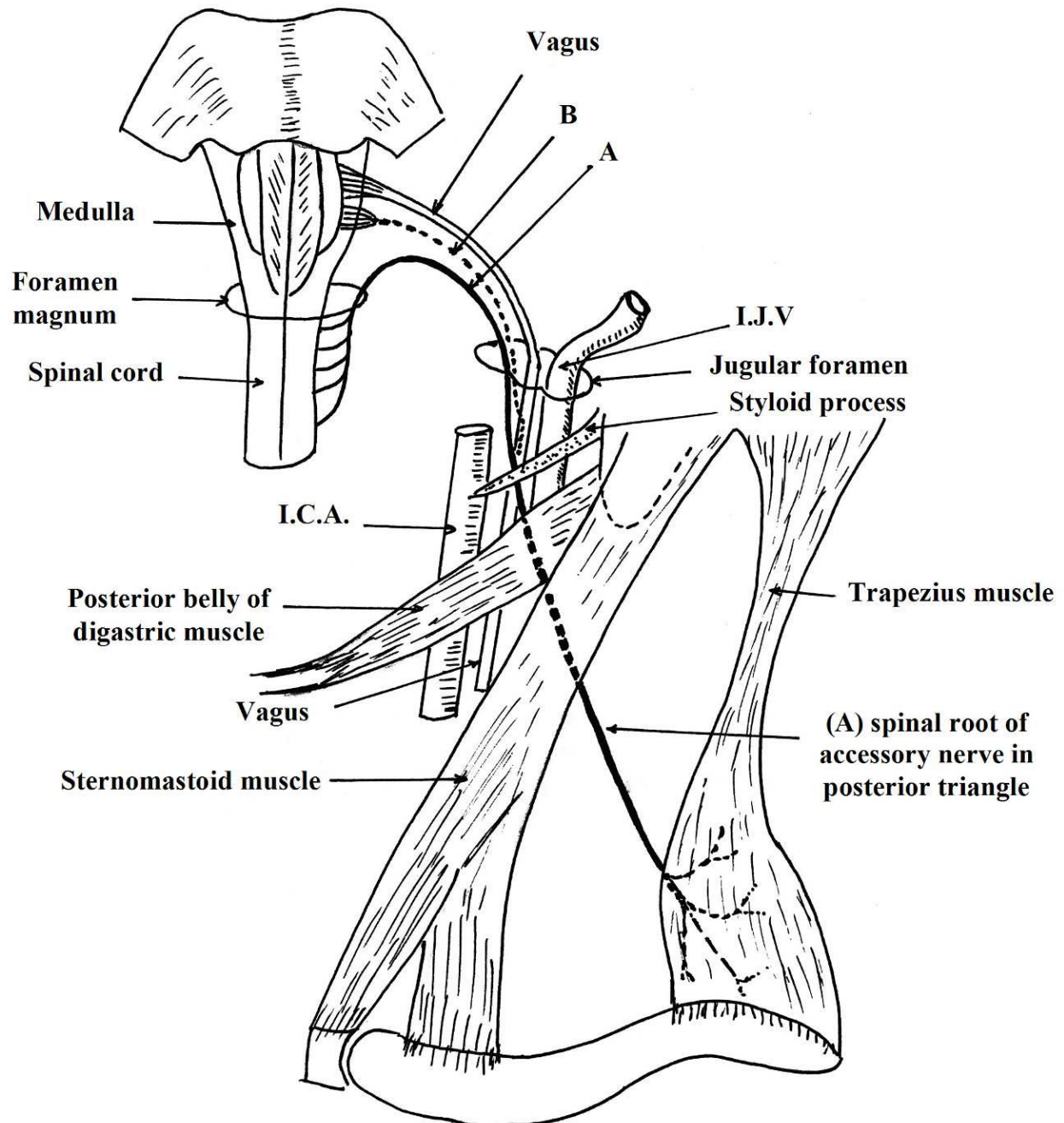
It arises from medulla oblongata from the lower part of nucleus ambiguus. It leaves the skull via jugular foramen where it joins the vagus nerve and distributes with its pharyngeal and laryngeal branches to supply:

- Muscles of larynx.
- Muscles of pharynx except stylopharyngeus.
- Muscles of palate except tensor palati.
- The 2 roots join each other in the jugular foramen and they separate below it.

Accessory Nerve

Two roots:

- A) Spinal root
- B) Cranial root



applied anatomy:

Injury of spinal part of accessory nerve:

Causes:

Because it is subcutaneous nerve it can be injured by superficial cut wound or by lymph node biopsy.

The injured nerve leads to weakness and atrophy of sternomastoid and trapezius muscles leading to impairment rotatory movement of neck and face to the opposite side.

Hypoglossal Nerve (12th cranial nerve)

It arises from the medulla from the hypoglossal nucleus and leaves the skull via hypoglossal canal.

It descends deep to posterior belly of digastric to enter the carotid triangle where it lies between IJV and ICA, then it curves forwards crossing 3 arteries; ICA, ECA and the loop of lingual artery. Then, it passes upwards deep to digastric muscle to enter the digastric triangle where it lies on the lateral surface of hyoglossus muscle below submandibular duct and deep part of submandibular gland. It finally divides into many branches on the surface of genioglossus muscle. Just below the skull it is joined by a branch from C1 which distributes with some of its branches.

Branches:

- **Four branches:** From the hypoglossal nerve itself to the following muscles:

1. Hyoglossus. 2. Styloglossus. 3. Genioglossus. 4. Intrinsic muscles of tongue.

- **Four branches from C1:**

1- Meningeal branch (to dura mater).

2- Descendens hypoglossi (shares in formation of ansa cervicalis).

3- Branch to thyrohyoid.

4- Branch to geniohyoid.

So it is a purely motor nerve which supplies all intrinsic and extrinsic muscles of the tongue except the palatoglossus muscle.

Applied anatomy:

Injury of hypoglossal nerve characterized by:

1- Hemiatrophy of the affected side of the tongue.

2- On protrusion of the tongue, it deviates towards the affected side.

Ansa Cervicalis

A nervous loop embedded in the anterior wall of carotid sheath & consists of 2 roots:

A- Superior root (*descendens hypoglossi*):

It is a branch from C1 which joins the hypoglossal nerve then descends in front of ICA and CCA in the anterior wall of carotid sheath.

B- Inferior root (*descendens cervicalis*):

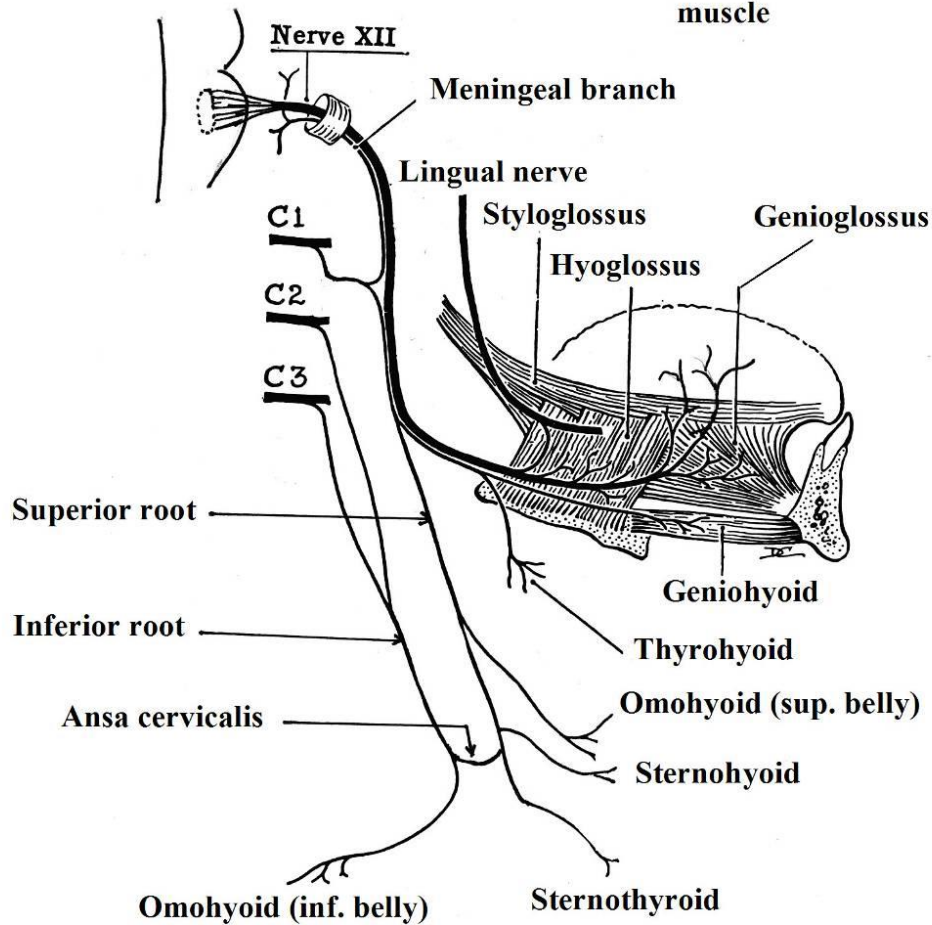
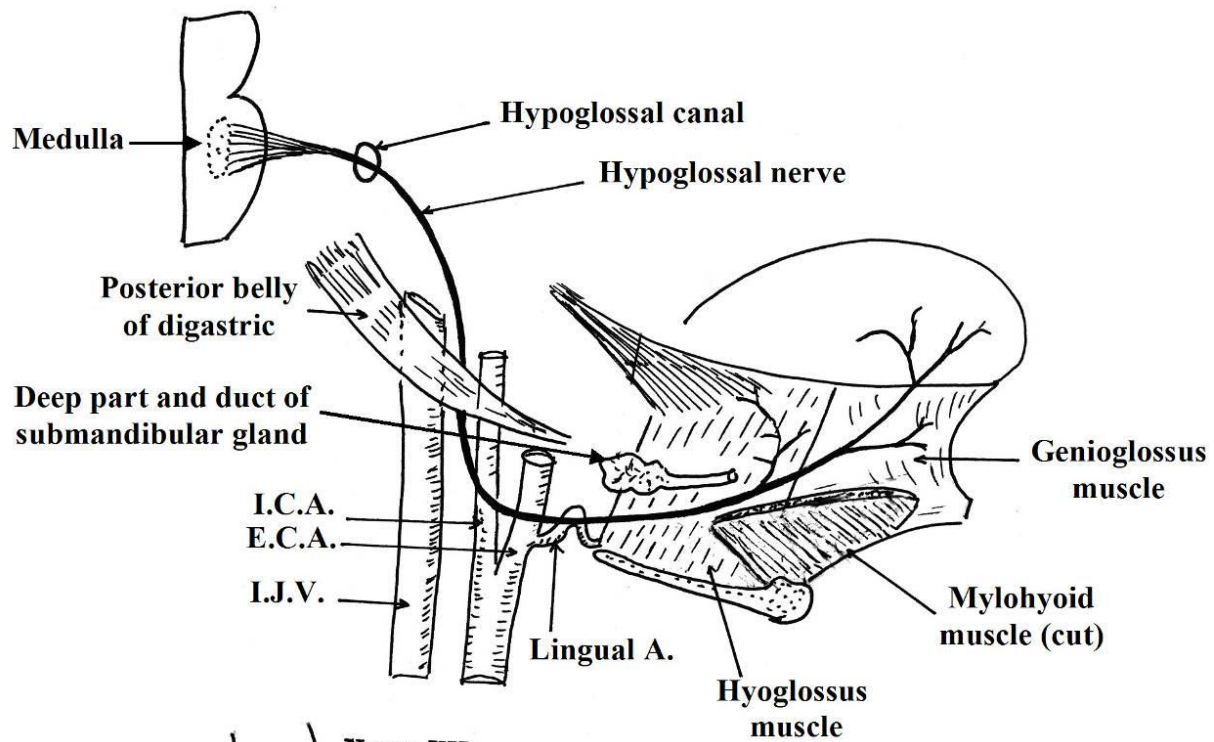
It arises from C2 and C3 and descends to join the superior root.

Branches: It supplies 3 infrahyoid muscles as follow:

1- Sternohyoid. **2-** Sternothyroid. **3-** Inferior belly of omohyoid.

All these muscles are supplied by branches from the ansa cervicalis itself.

The superior belly of omohyoid is supplied by separate branch from the superior root of the ansa.



Branches of Hypoglossal Nerve

Cervical Part of Sympathetic Trunk

Ganglionated nerve trunk embedded in the posterior wall of the carotid sheath and carries 3 ganglia. It lies behind CCA and ICA and in front of inferior thyroid artery, prevertebral muscles, fascia and transverse processes of cervical vertebrae.

	Superior ganglion	Middle ganglion	Inferior ganglion
Position	Lies in front of the transverse processes of C2 and C3 vertebrae and behind ICA (2-3 cm long)	Lies in front of the transverse process of C6 vertebra, and inferior thyroid artery (the smallest ganglion)	Lies in front of the transverse process of C7 vertebra behind vertebral artery (single or fused with 1 st thoracic ganglion)
Branches	1. Cardiac branch to the heart 2. Vascular branches forming plexus around ECA and ICA 3. Communicating branches to upper 4 cervical and lower 4 cranial nerves 4. Visceral branches: pharyngeal branch to pharyngeal plexus	1. Cardiac branch to the heart 2. Vascular branches forming plexus around inferior thyroid artery 3. Communicating branches to C5 and C6 nerves 4. Visceral branches to trachea and oesophagus	1. Cardiac branch to the heart 2. Vascular branches forming plexus around vertebral and subclavian arteries 3. Communicating branches to C7 and C8 nerves 4. Branch forming ansa subclavia with a branch coming from the middle ganglion